Oral Cavity
Tongue & Palate

first part

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Oral Cavity (Mouth).

1) Tongue.

2) Palate.

- **Oral Cavity (Mouth)**
  - A. Vestibule.
  - B. Oral Cavity Proper
  - C. Floor of The Mouth

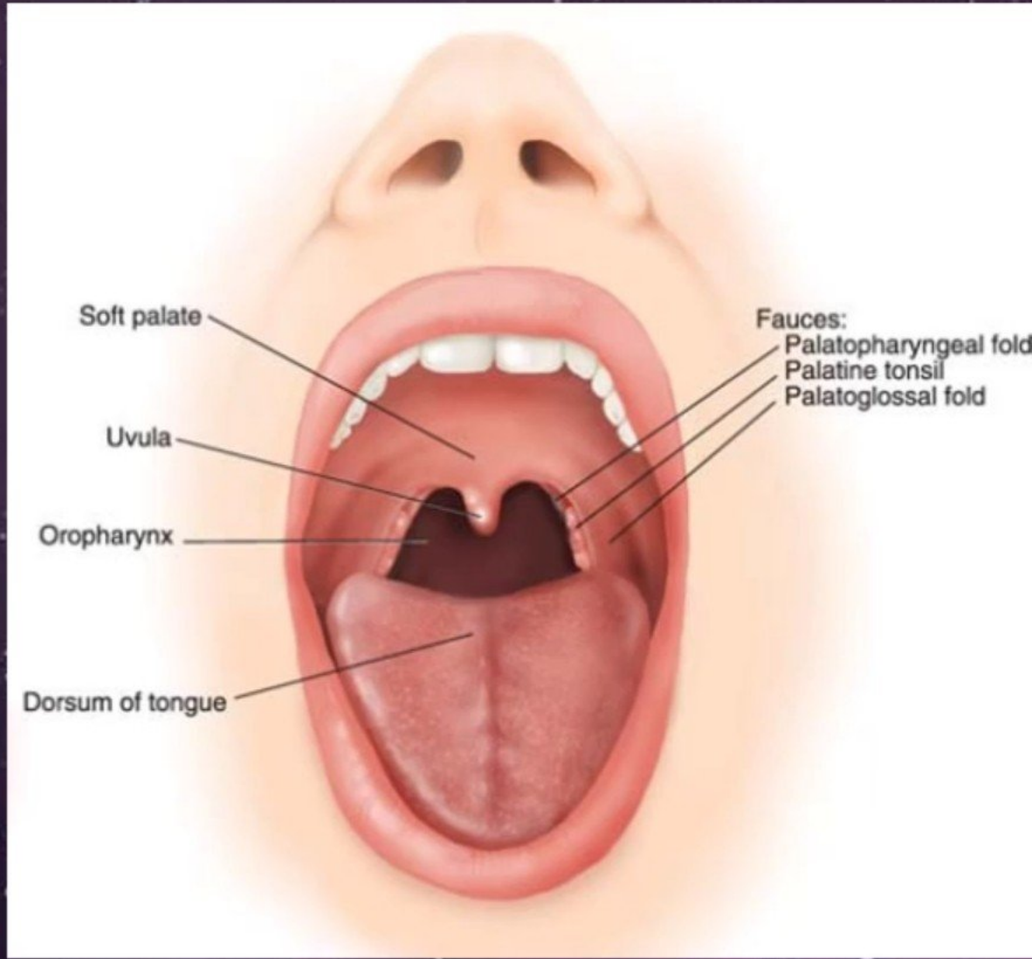
1) Nerve Supply of Oral Cavity
2) Nerve Supply Of Gum
3) Innervation of gingivae and teeth
4) Teeth

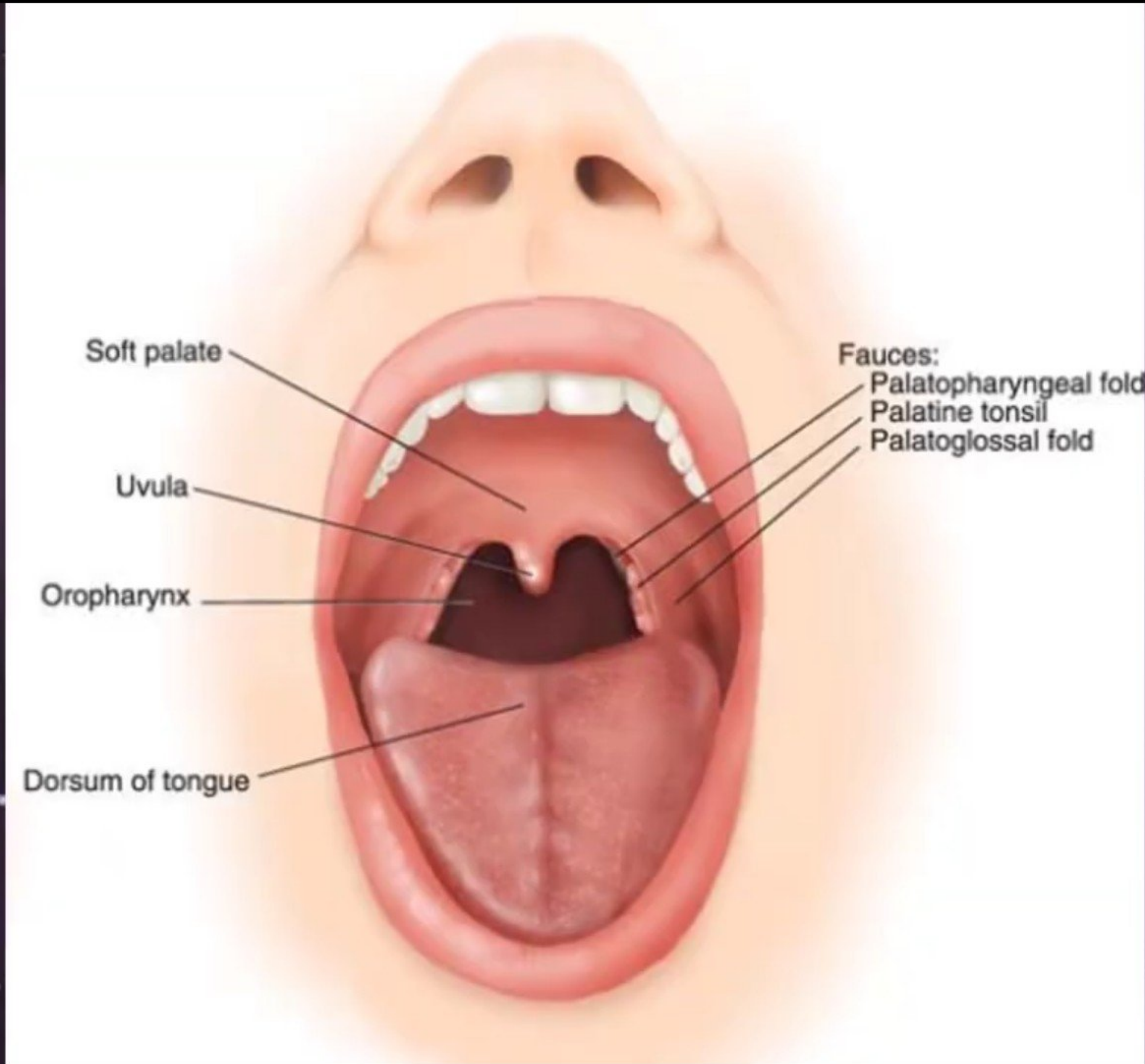
1) Tongue

1) Dorsal Surface of the tongue.
2) Ventral Surface of the tongue.
3) Muscles of the tongue.
   - A. Intrinsic Muscles
   - B. Extrinsic Muscles
   - C. Movements of the tongue
4) Sensory Nerve Supply
5) Motor Nerve Supply.
6) Blood Supply.
7) Lymphatic Drainage.
8) Functions.
9) Clinical note.

Oral Cavity (Mouth)

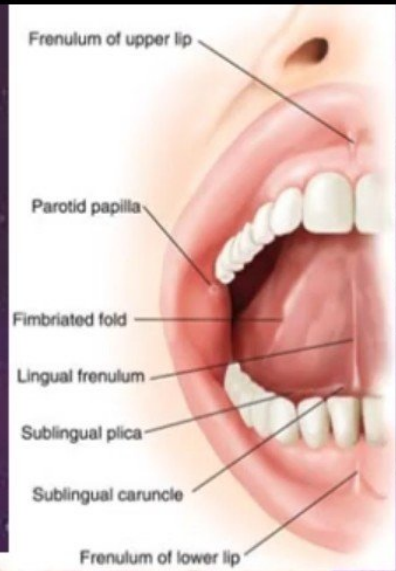
- Extends from the lips to the oropharyngeal isthmus
  - The oropharyngeal isthmus:
    - Is the junction of mouth and pharynx.
    - Is bounded:
      - Above by the soft palate and the palatoglossal folds
      - Below by the dorsum of the tongue
- Subdivided into Vestibule & Oral cavity proper

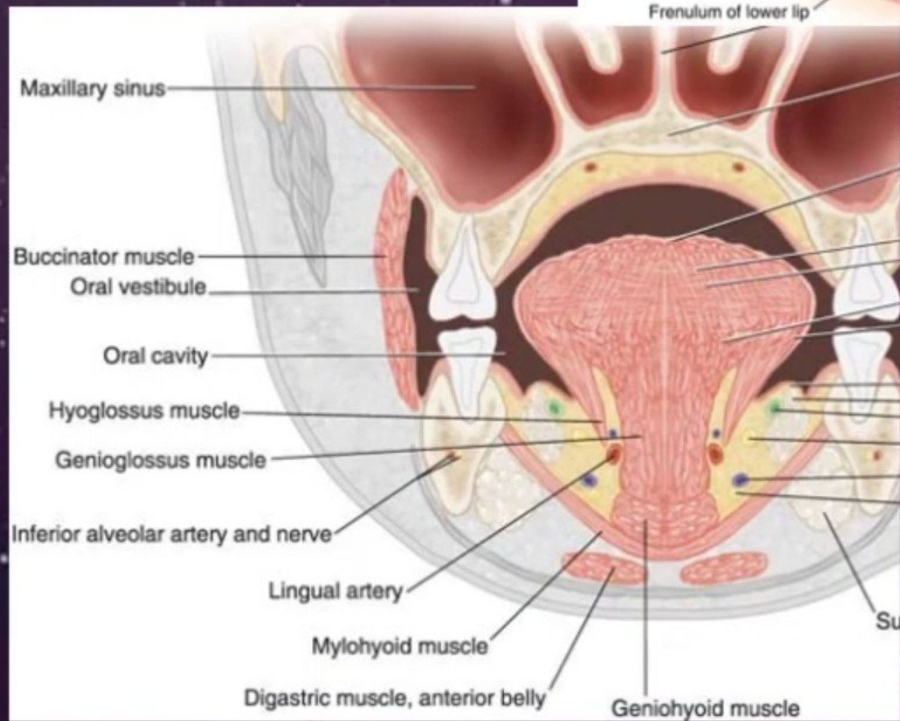


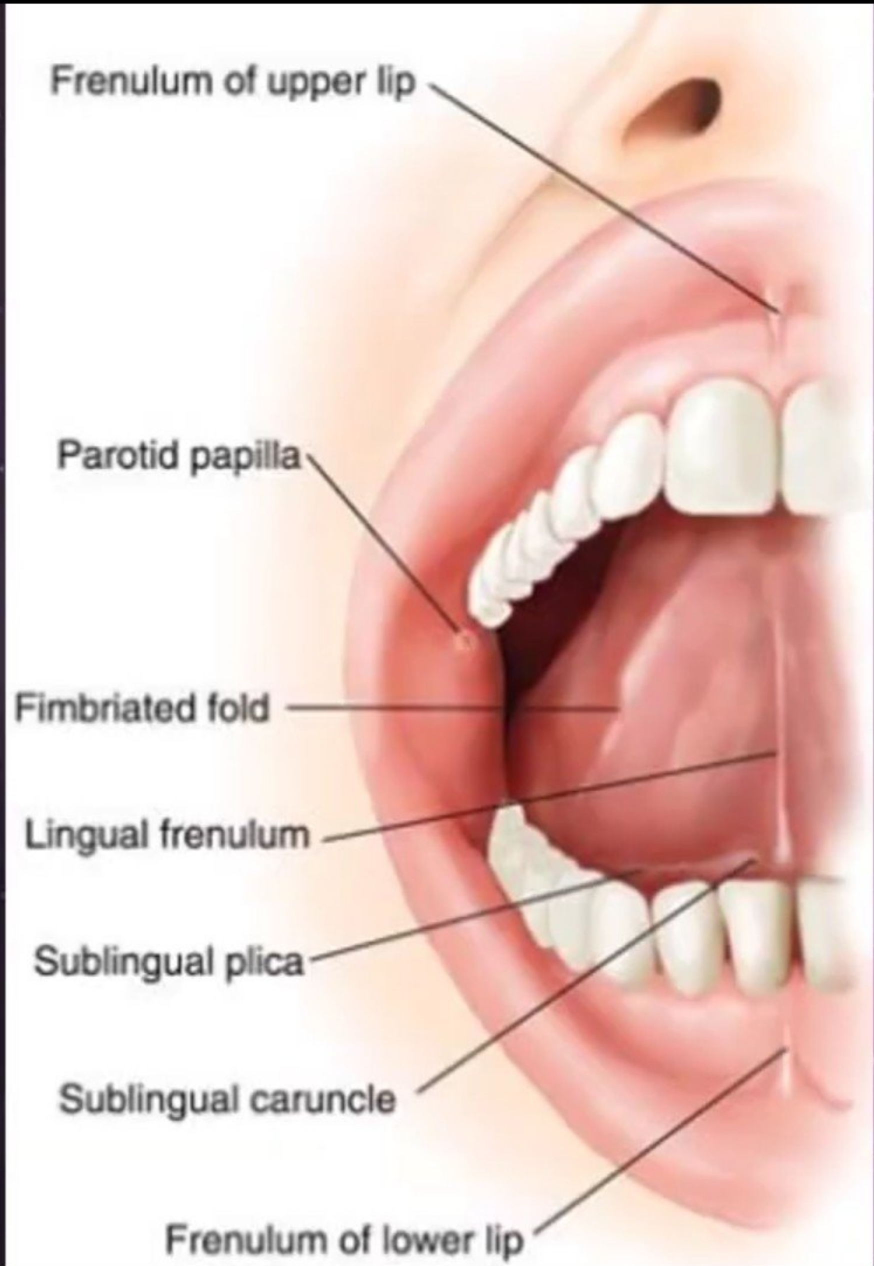

Soft palate
Uvula
Oropharynx
Dorsum of tongue
Fauces:
Palatopharyngeal fold
Palatine tonsil
Palatoglossal fold

Vestibule

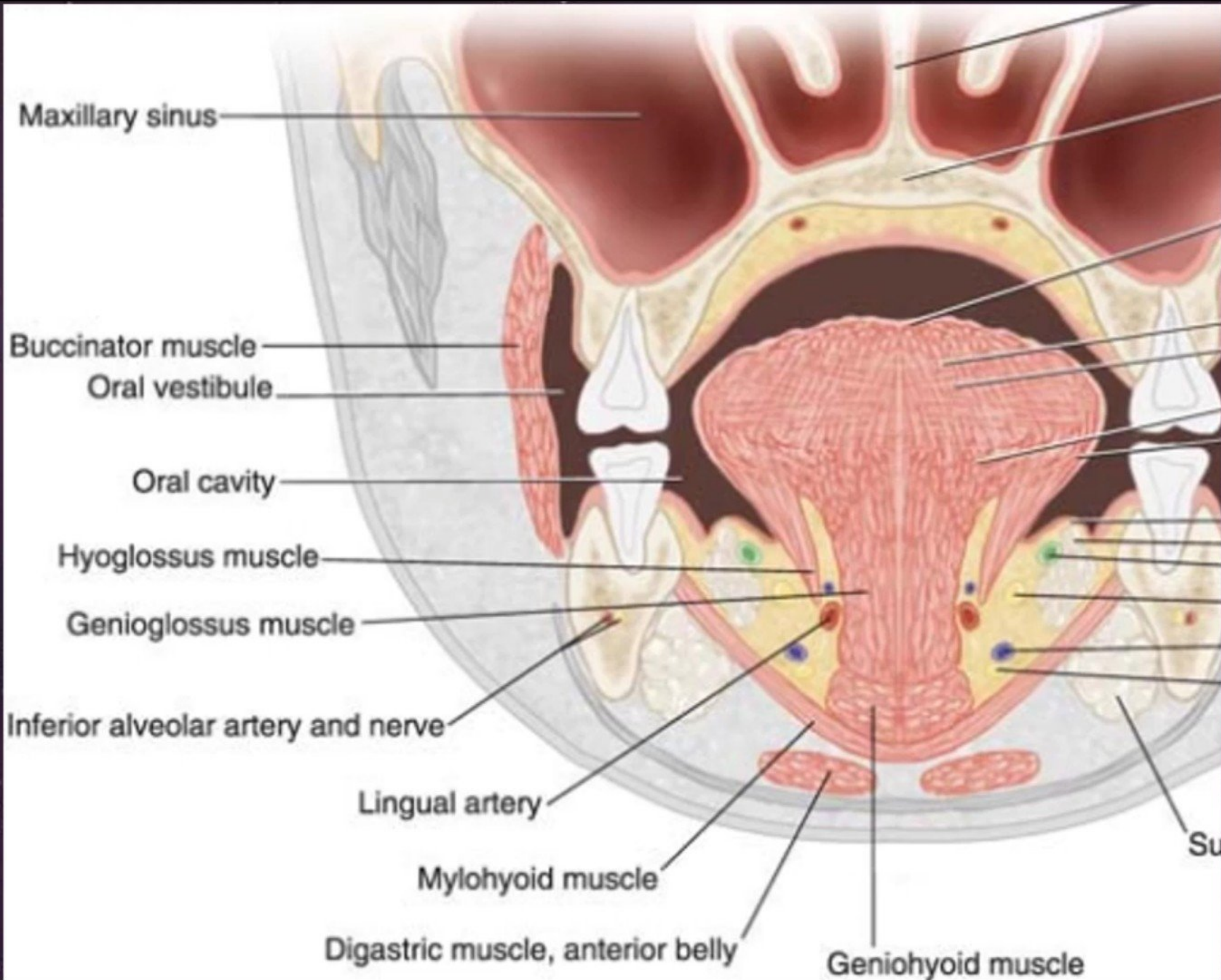
- Slitlike space between the cheeks and the gums
- Communicates with the exterior through the oral fissure
- When the jaws are closed, communicates with the oral cavity proper behind the 3rd molar tooth on each side
- Superiorly and inferiorly limited by the reflection of mucous membrane from lips and cheek onto the gums





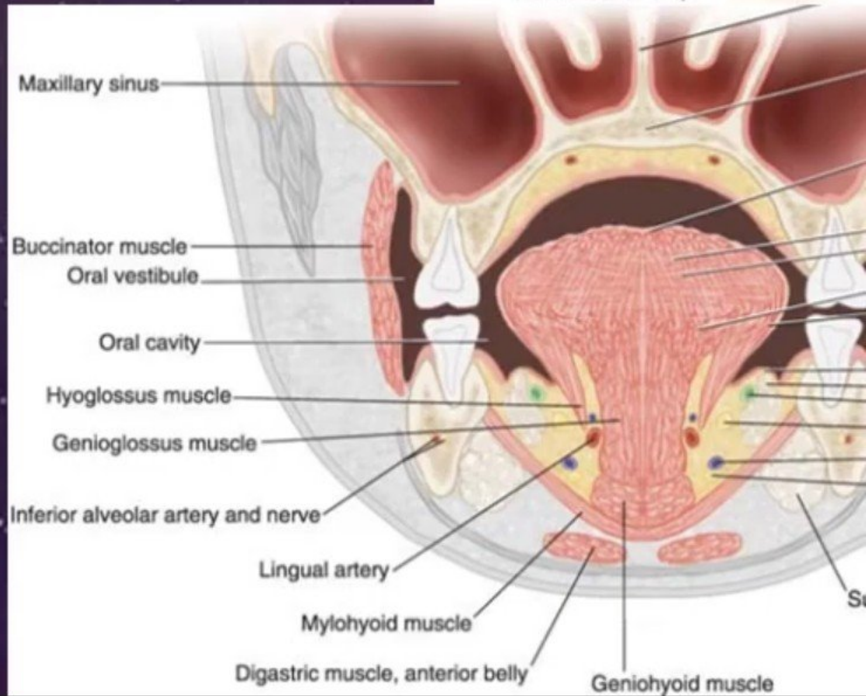


Frenulum of upper lip
Parotid papilla
Fimbriated fold
Lingual frenulum
Sublingual plica
Sublingual caruncle
Frenulum of lower lip


Maxillary sinus
Buccinator muscle
Oral vestibule
Oral cavity
Hyoglossus muscle
Genioglossus muscle
Inferior alveolar artery and nerve
Lingual artery
Mylohyoid muscle
Digastric muscle, anterior belly
Geniohyoid muscle

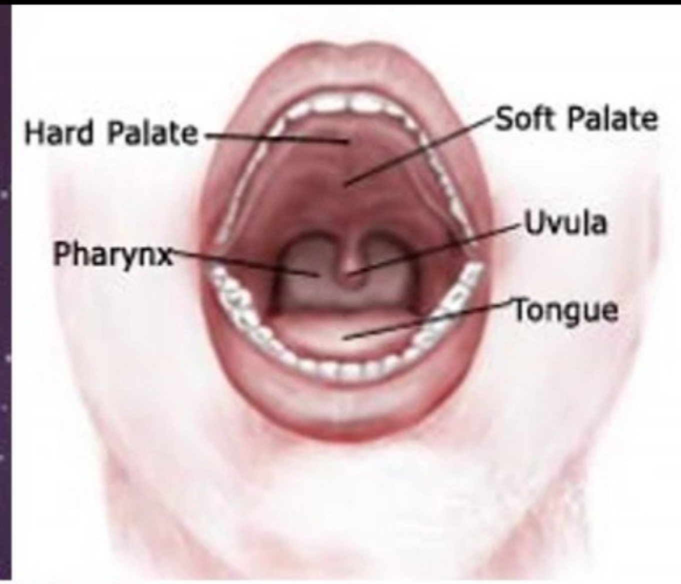
Vestibule

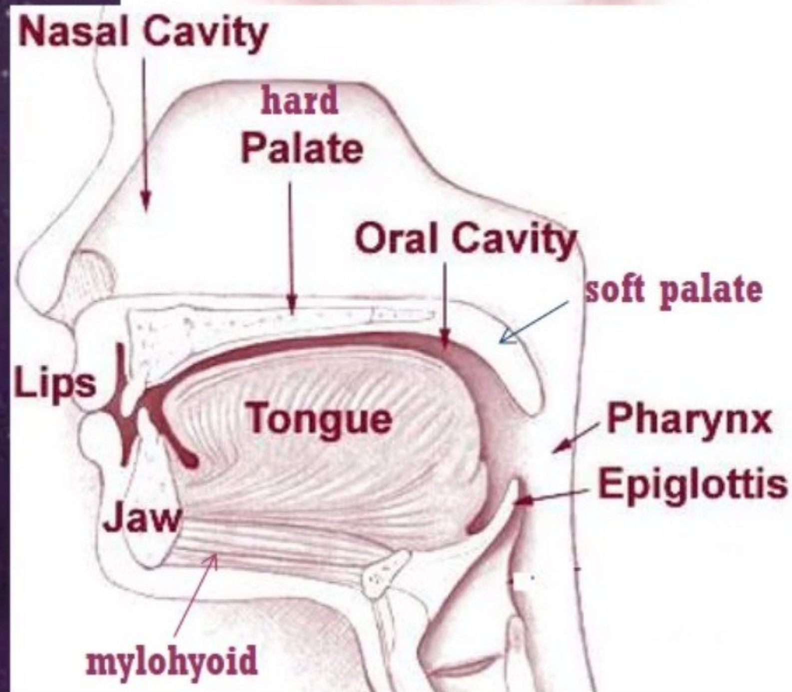
- The lateral wall of the vestibule is formed by the cheek
  - The cheek is composed of Buccinator muscle, covered laterally by the skin & medially by the mucous membrane
- A small papilla on the mucosa opposite the upper 2nd molar tooth marks the opening of the duct of the parotid gland

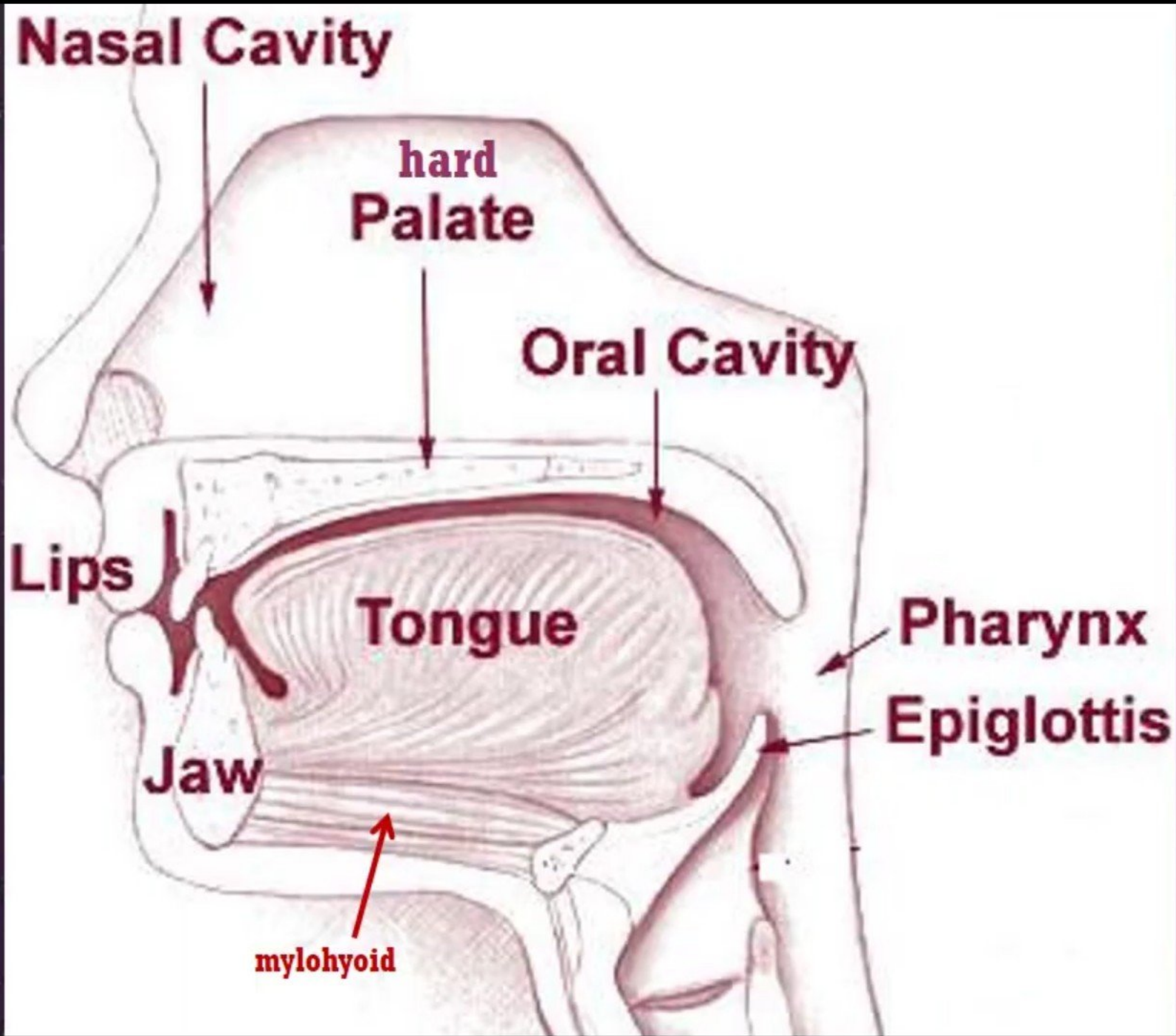


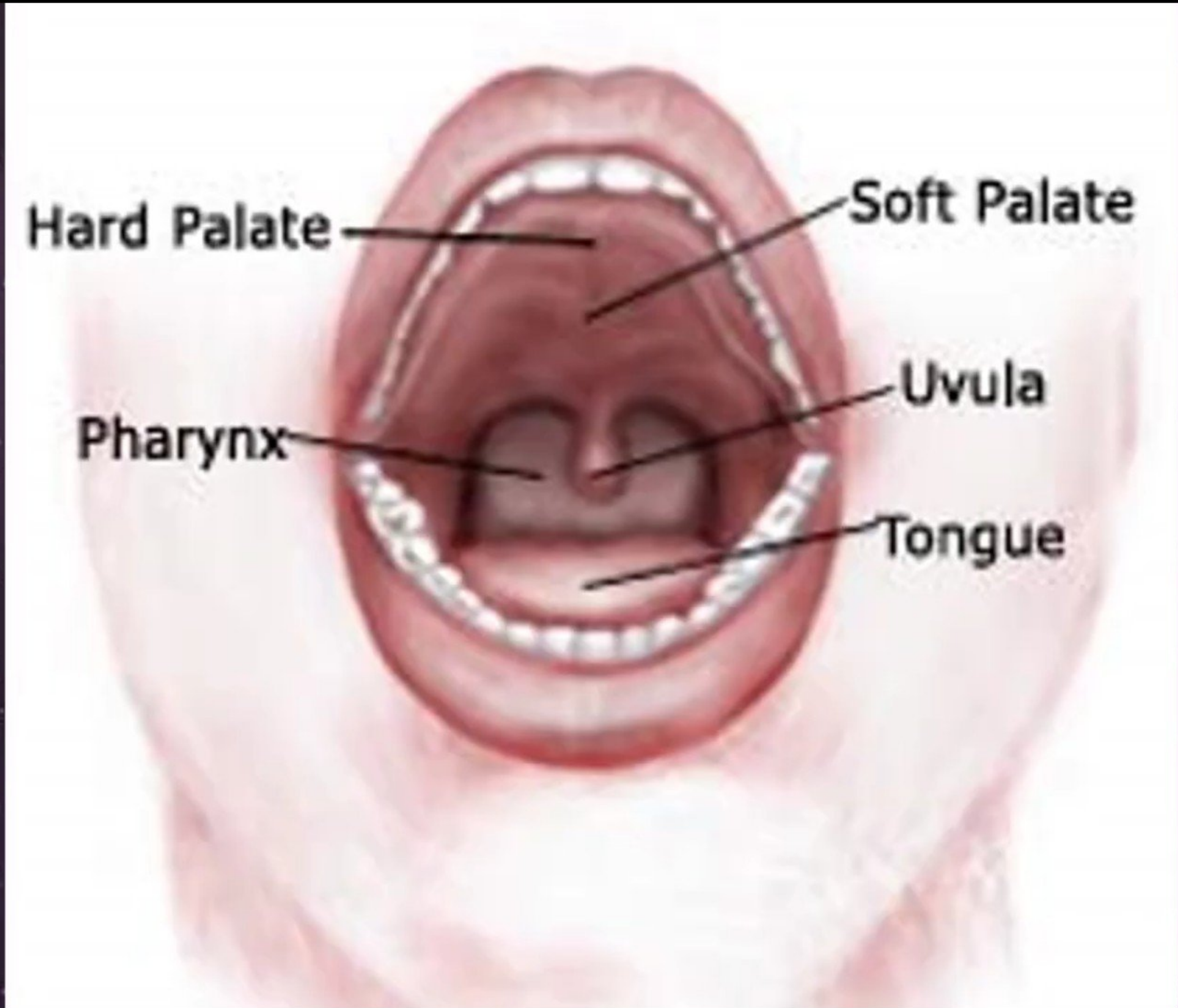
Oral Cavity Proper

- It is the cavity within the alveolar margins of the maxillae and the mandible
- Its Roof is formed by the hard palate anteriorly and the soft palate posteriorly.
- Its Floor is formed by the mylohyoid muscle. The anterior 2/3rd of the tongue lies on the floor.



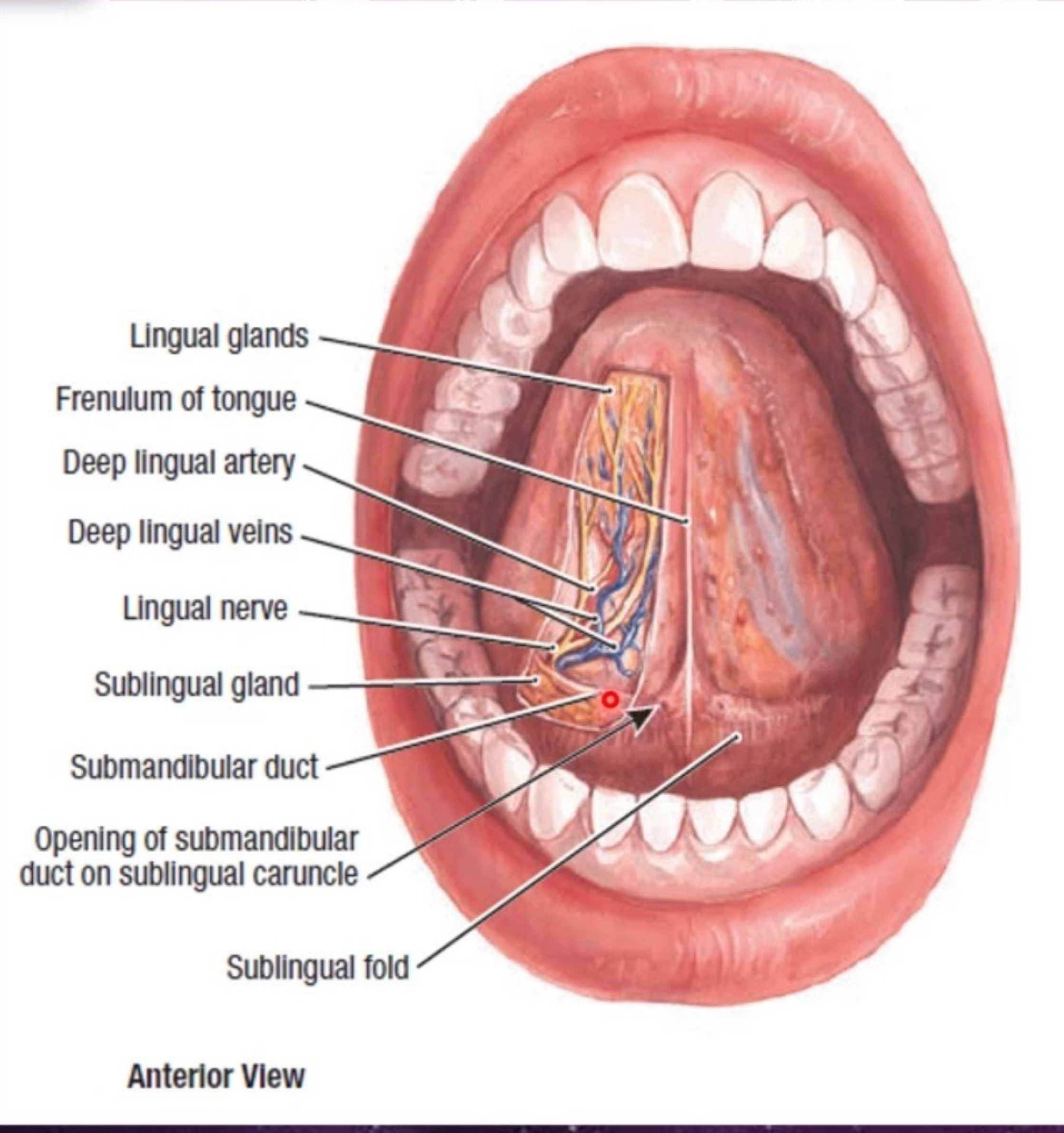


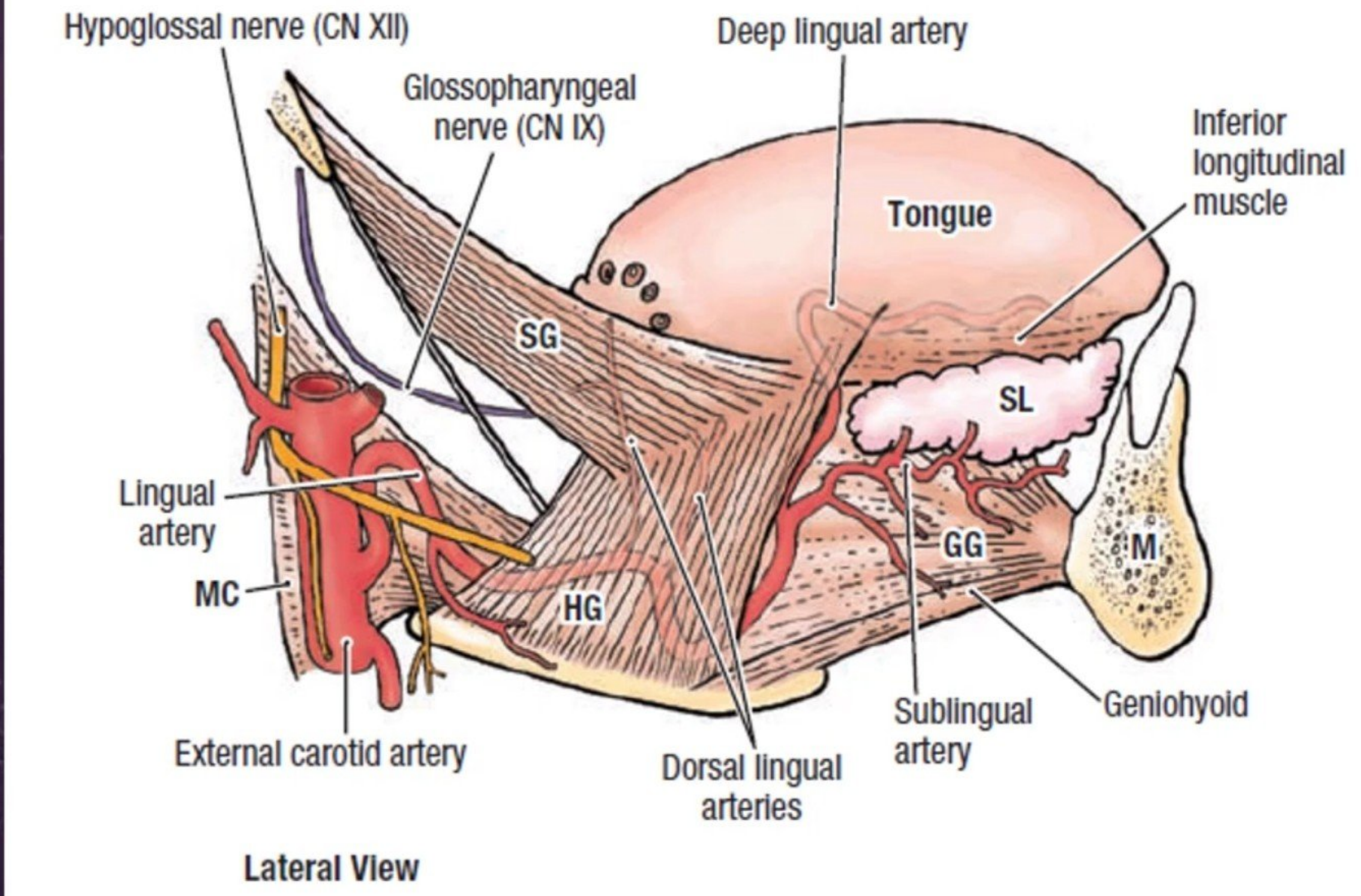

Nasal Cavity
hard
Palate
Oral Cavity
Lips
Tongue
Pharynx
Epiglottis
Jaw
mylohyoid

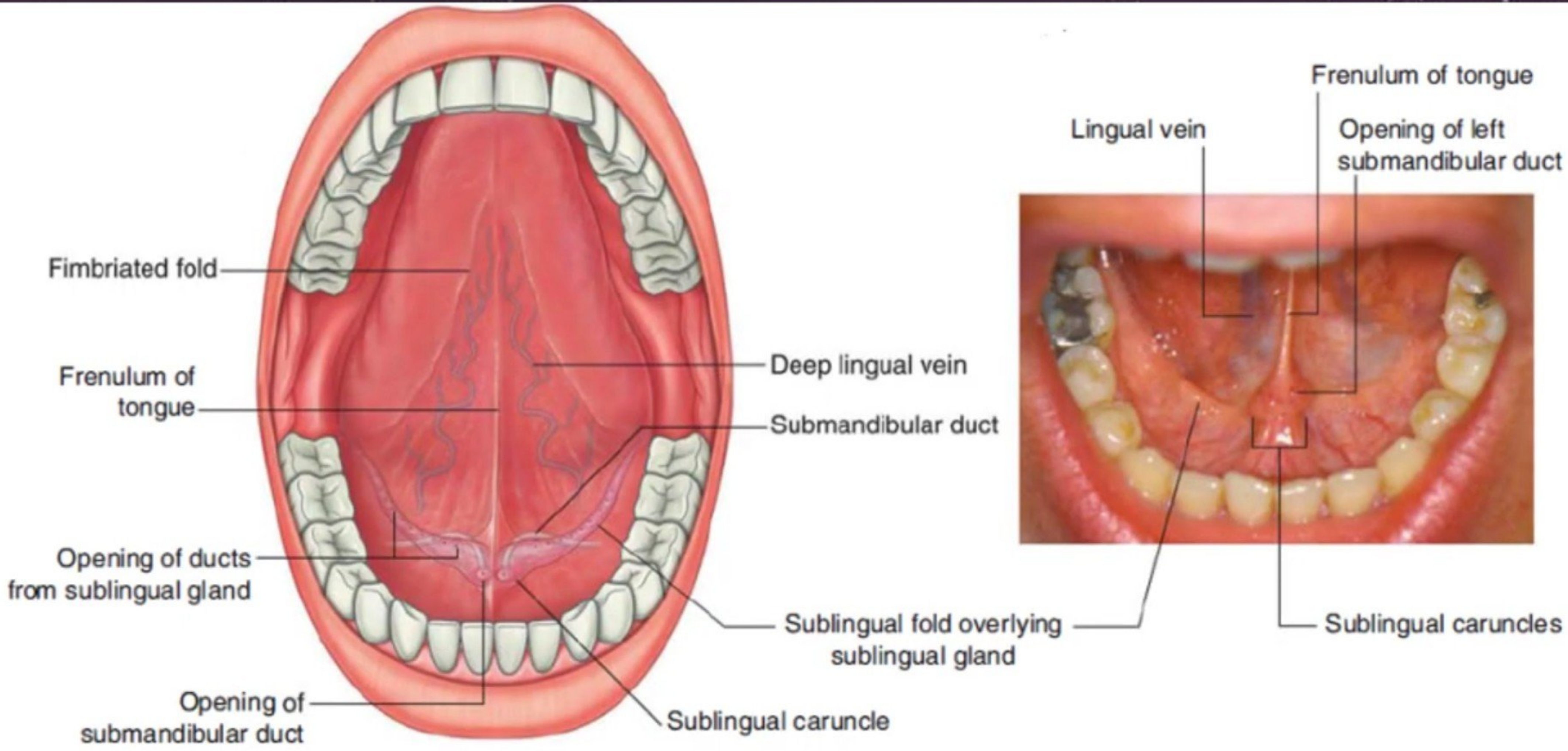

Hard Palate
Soft Palate
Uvula
Pharynx
Tongue

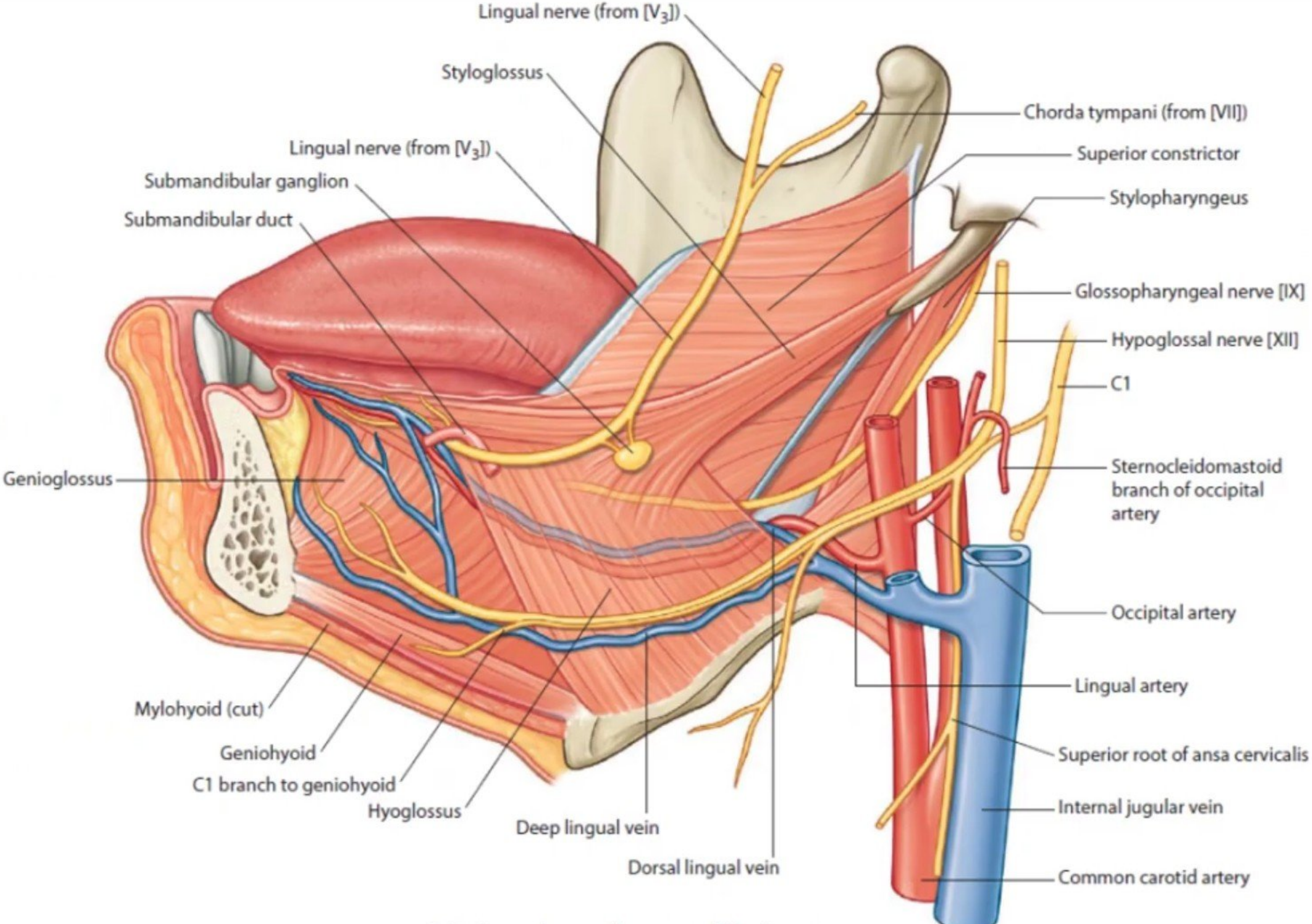
Floor of The Mouth

- Covered with mucous membrane
- In the midline, a mucosal fold, the frenulum, connects the tongue to the floor of the mouth
- On each side of frenulum a small papilla has the opening of the duct of the submandibular gland
- A rounded ridge extending backward & laterally from the papilla is produced by the sublingual gland

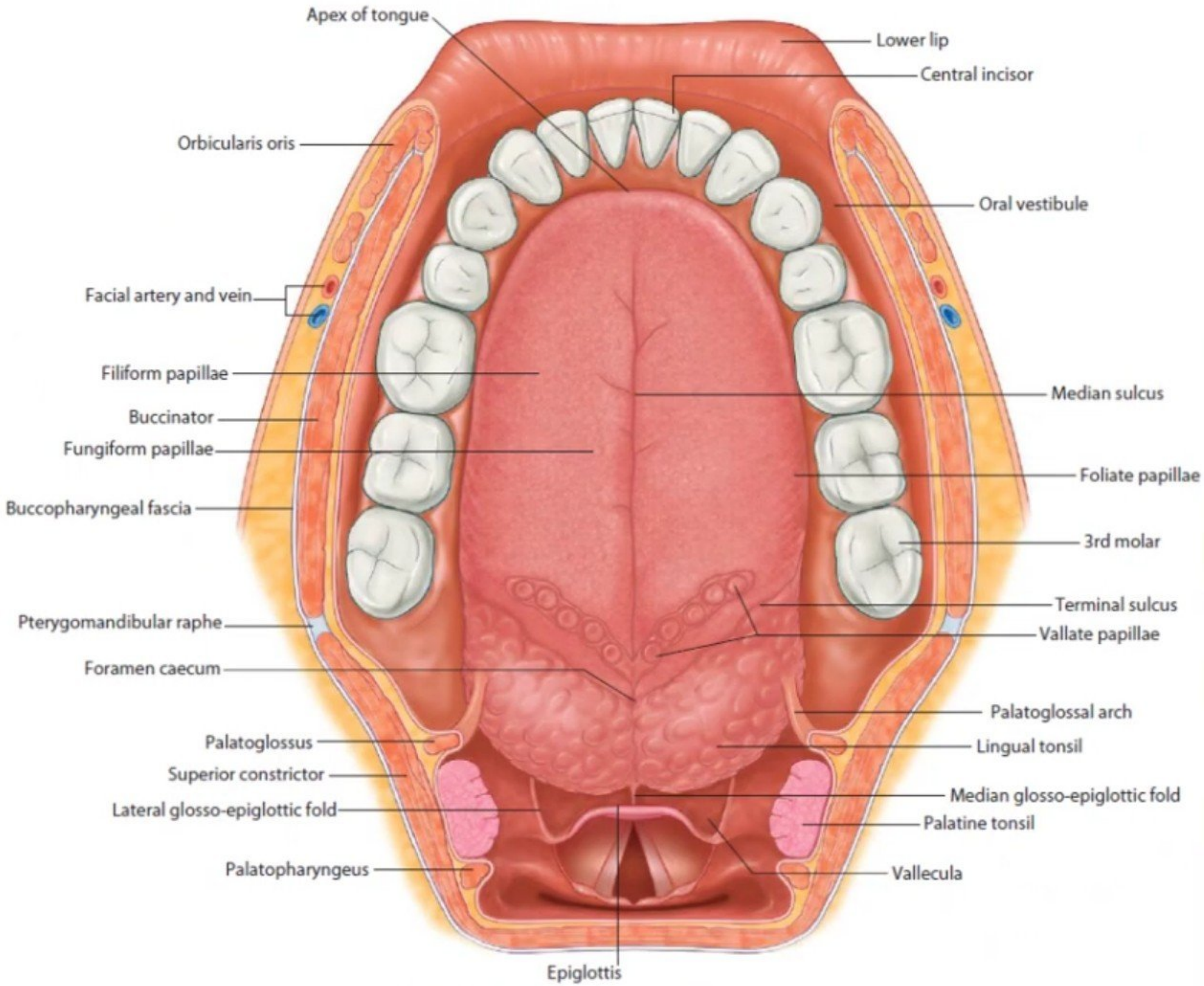



Hypoglossal nerve (CN XII)
Glossopharyngeal nerve (CN IX)
Deep lingual artery
Inferior longitudinal muscle
Tongue
SG
SL
Lingual artery
MC
HG
GG
M
External carotid artery
Dorsal lingual arteries
Sublingual artery
Geniohyoid
Lateral View


Fimbriated fold
Frenulum of tongue
Opening of ducts from sublingual gland
Opening of submandibular duct
Deep lingual vein
Submandibular duct
Sublingual fold overlying sublingual gland
Sublingual caruncle
Frenulum of tongue
Lingual vein
Opening of left submandibular duct
Sublingual caruncles



Arteries, veins, and nerves of the tongue



Horizontal section showing superior view of tongue

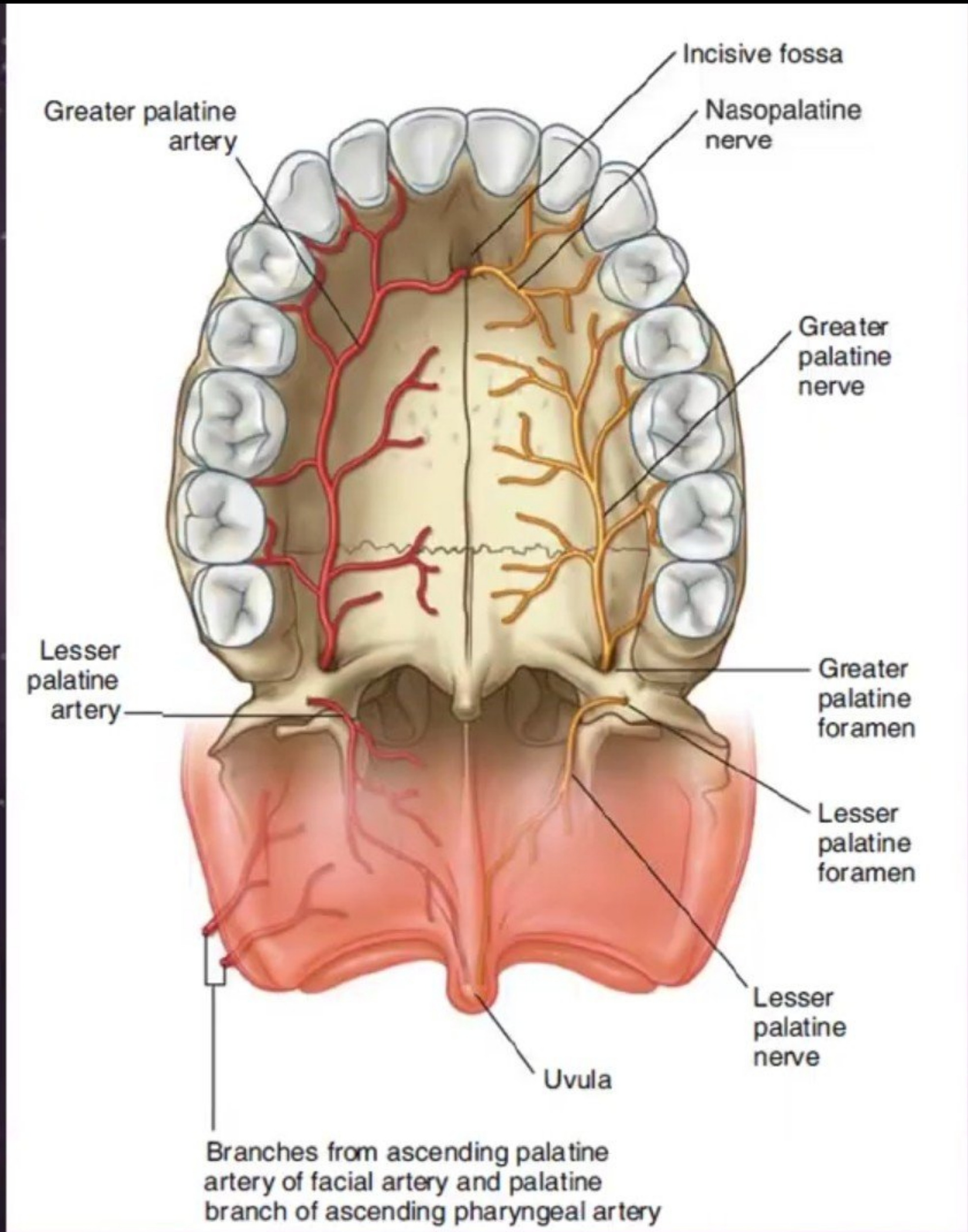
Nerve Supply of Oral Cavity

- **Sensory**
  - Roof: by greater palatine and nasopalatine nerves (branches of maxillary nerve)
  - Floor: by lingual nerve (branch of mandibular nerve)
  - Cheek: by buccal nerve (branch of mandibular nerve)
- **Motor**
  - Muscle in the cheek (buccinator) and the lip (orbicularis oris) are supplied by the branches of the facial nerve

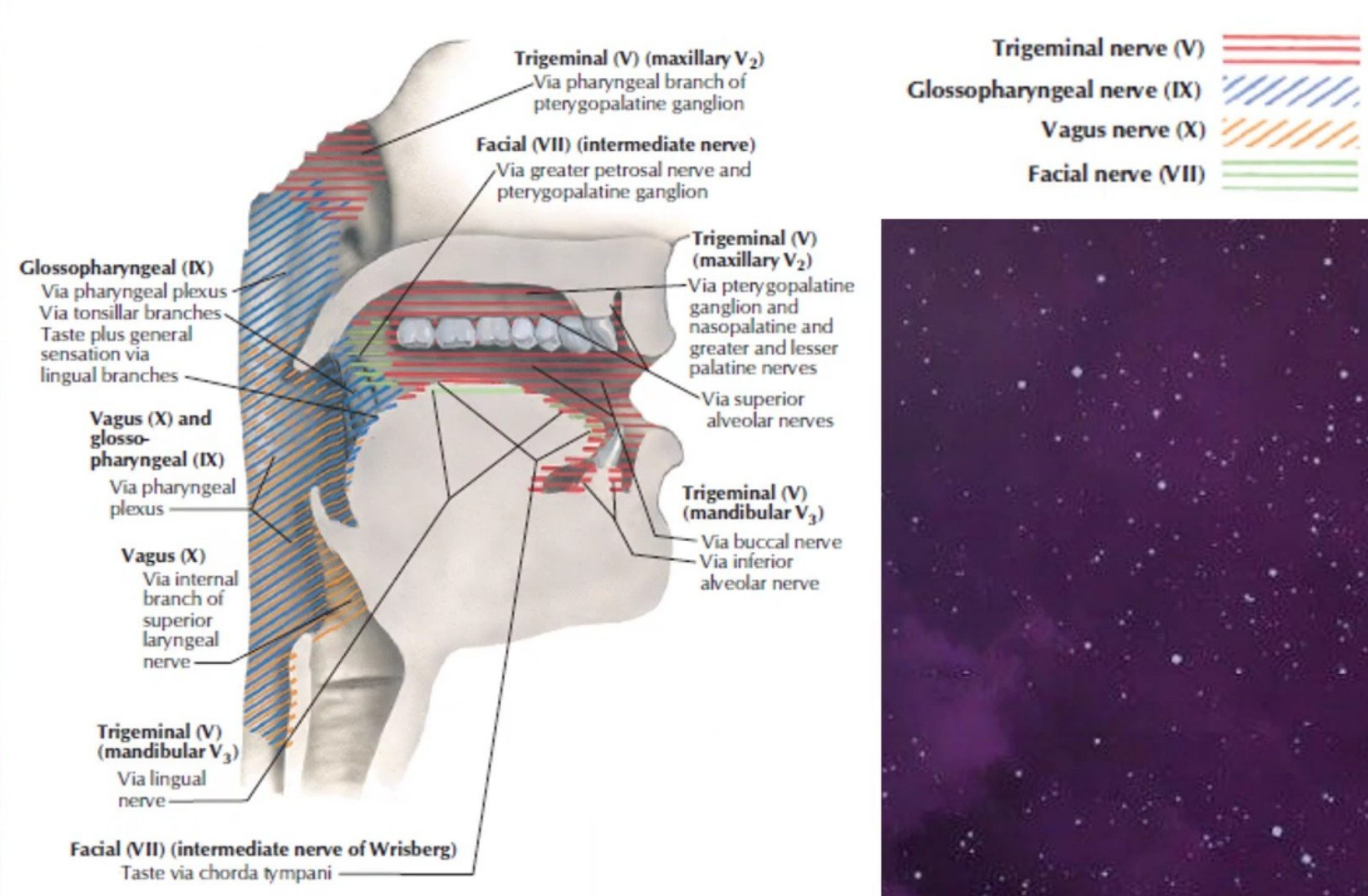
Nerve Supply Of Gum

- **The upper gums** are supplied by the *superior alveolar*, *greater palatine* and *nasopalatine nerves* (maxillary),
- **The lower gums** receive their innervation from the *inferior alveolar*, *buccal* and *lingual nerves* (mandibular).

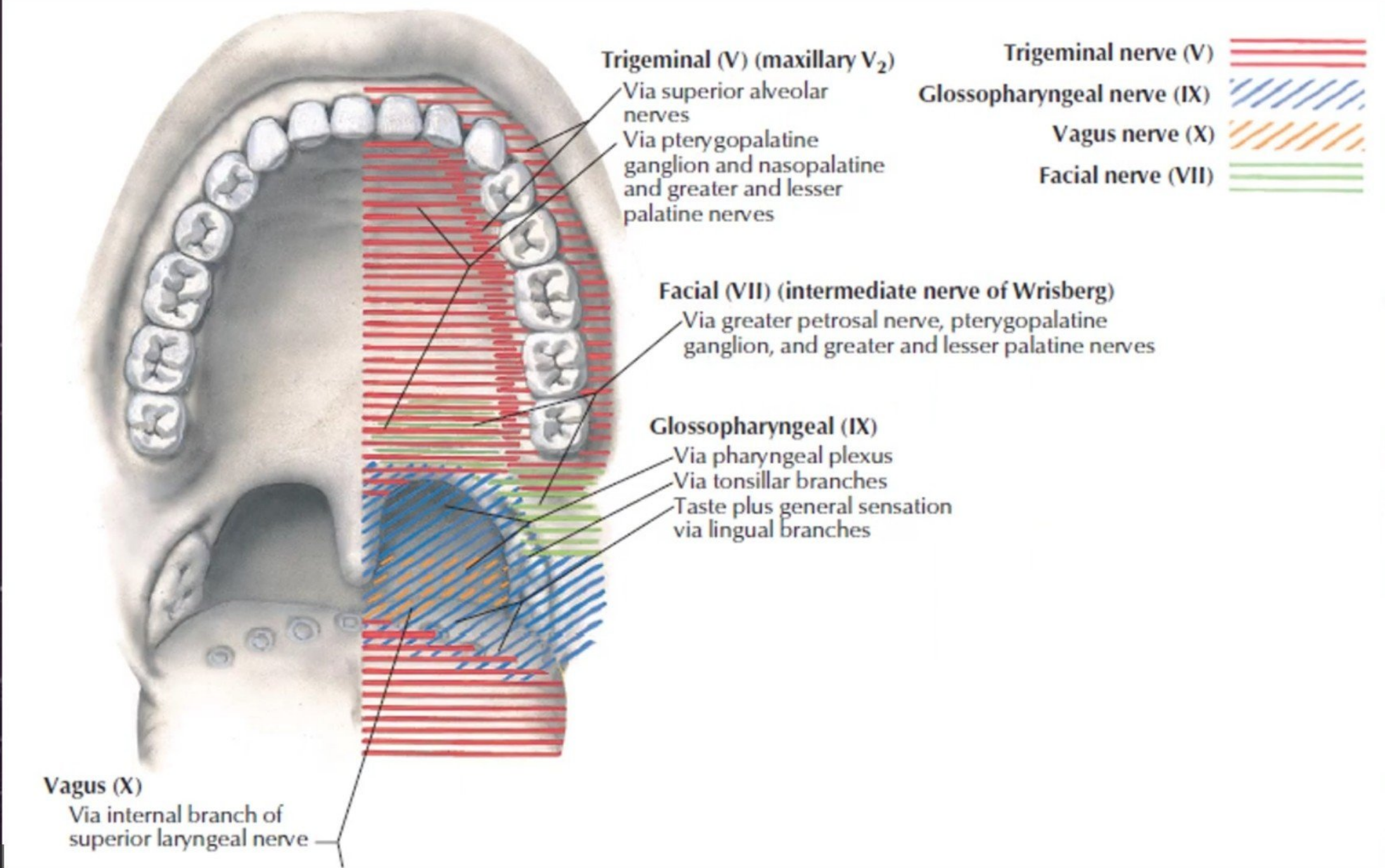
The buccal nerve does not usually innervate the upper gums.



Incisive fossa
Greater palatine artery
Nasopalatine nerve
Greater palatine nerve
Lesser palatine artery
Greater palatine foramen
Lesser palatine foramen
Lesser palatine nerve
Uvula
Branches from ascending palatine artery of facial artery and palatine branch of ascending pharyngeal artery

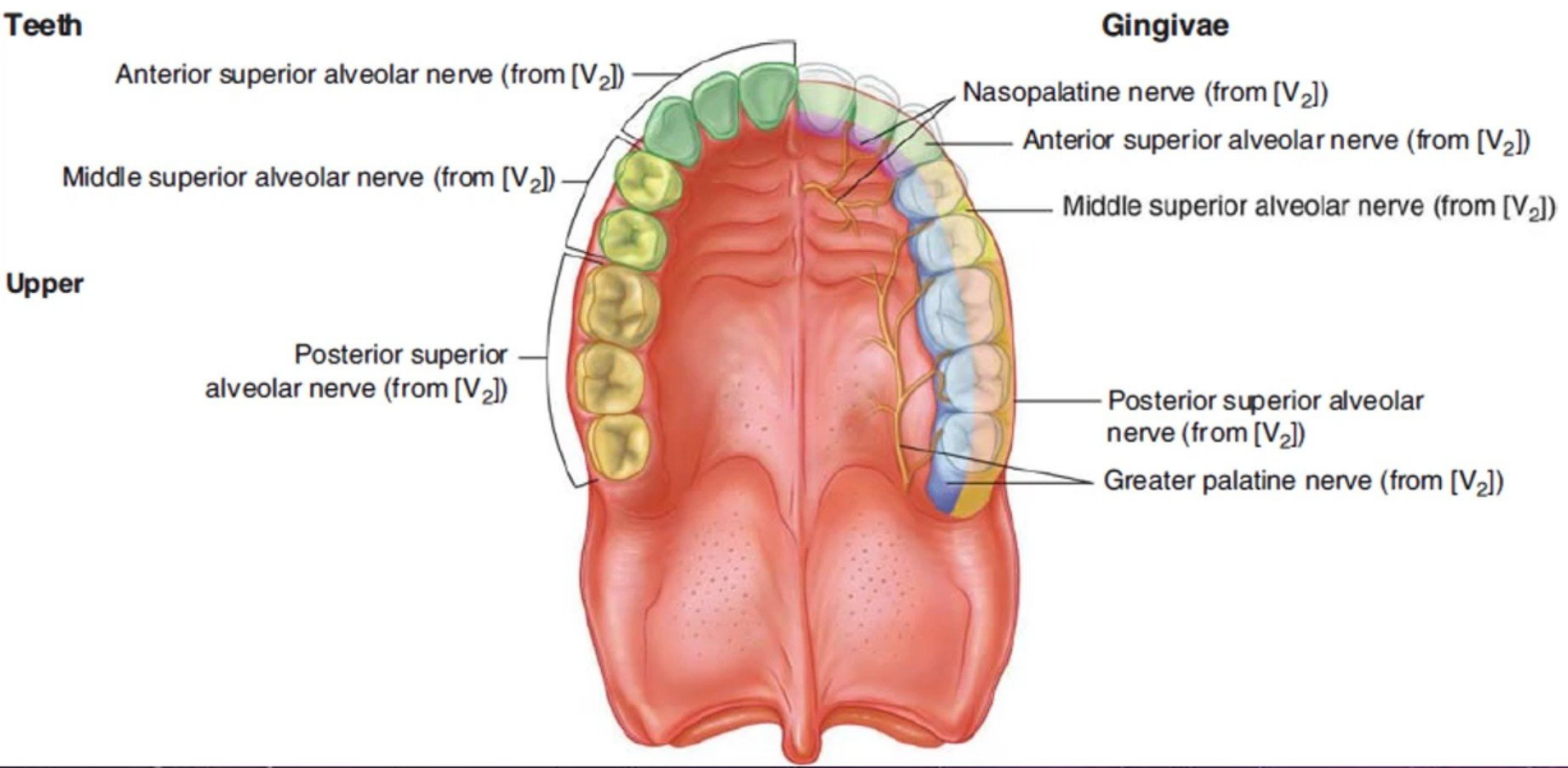


Trigeminal (V) (maxillary V_2)
Via pharyngeal branch of pterygopalatine ganglion
Facial (VII) (intermediate nerve)
Via greater petrosal nerve and pterygopalatine ganglion
Glossopharyngeal (IX)
Via pharyngeal plexus
Via tonsillar branches
Taste plus general sensation via lingual branches
Vagus (X) and glossopharyngeal (IX)
Via pharyngeal plexus
Vagus (X)
Via internal branch of superior laryngeal nerve
Trigeminal (V) (mandibular V_3)
Via lingual nerve
Facial (VII) (intermediate nerve of Wrisberg)
Taste via chorda tympani
Trigeminal (V) (maxillary V_2)
Via pterygopalatine ganglion and nasopalatine and greater and lesser palatine nerves
Via superior alveolar nerves
Trigeminal (V) (mandibular V_3)
Via buccal nerve
Via inferior alveolar nerve
Trigeminal nerve (V)
Glossopharyngeal nerve (IX)
Vagus nerve (X)
Facial nerve (VII)



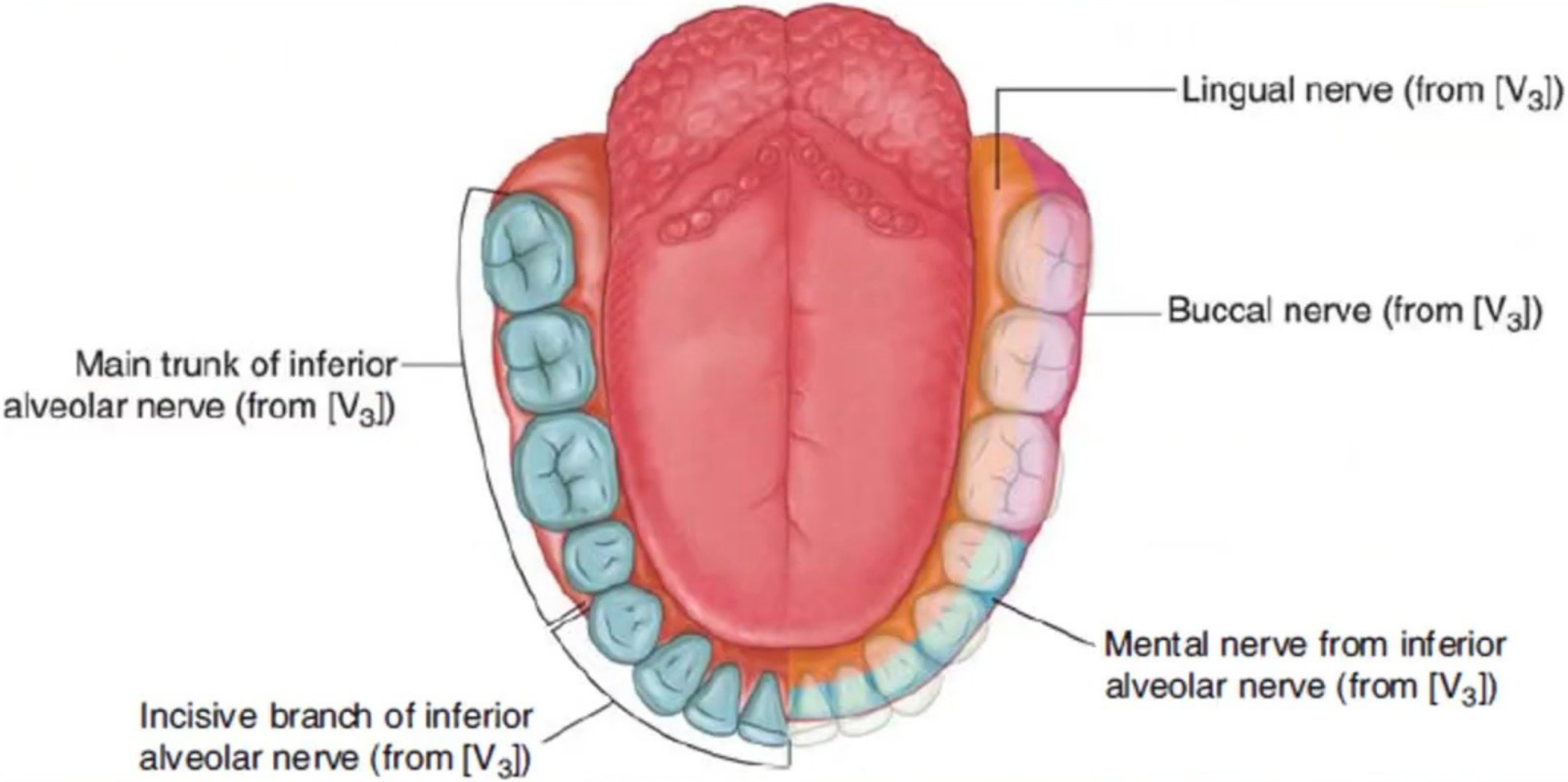
Trigeminal (V) (maxillary V_2)
Via superior alveolar nerves
Via pterygopalatine ganglion and nasopalatine and greater and lesser palatine nerves
Trigeminal nerve (V)
Glossopharyngeal nerve (IX)
Vagus nerve (X)
Facial nerve (VII)
Facial (VII) (intermediate nerve of Wrisberg)
Via greater petrosal nerve, pterygopalatine ganglion, and greater and lesser palatine nerves
Glossopharyngeal (IX)
Via pharyngeal plexus
Via tonsillar branches
Taste plus general sensation via lingual branches
Vagus (X)
Via internal branch of superior laryngeal nerve

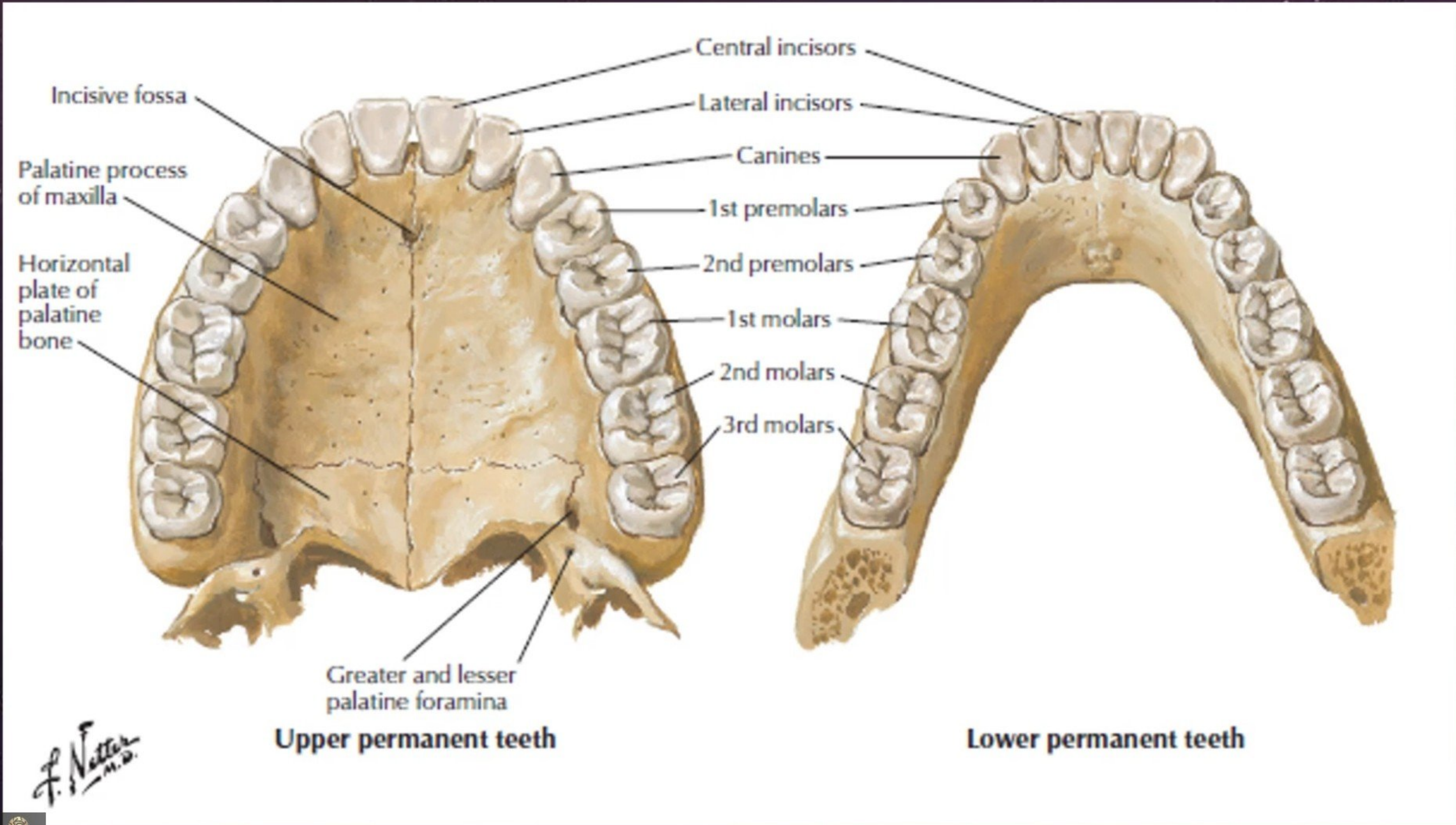
Innervation of gingivae and teeth

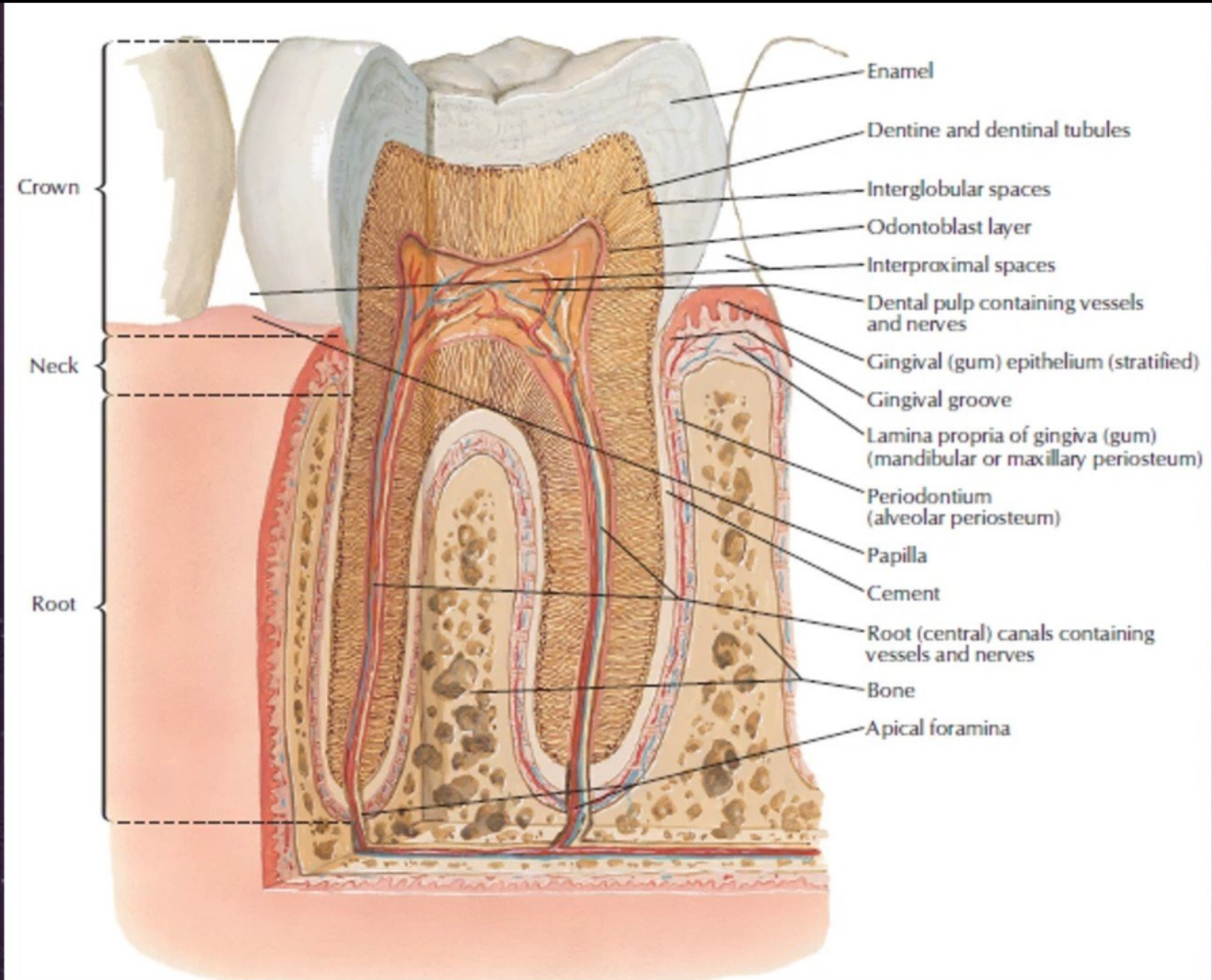


Innervation of gingivae and teeth

Lower

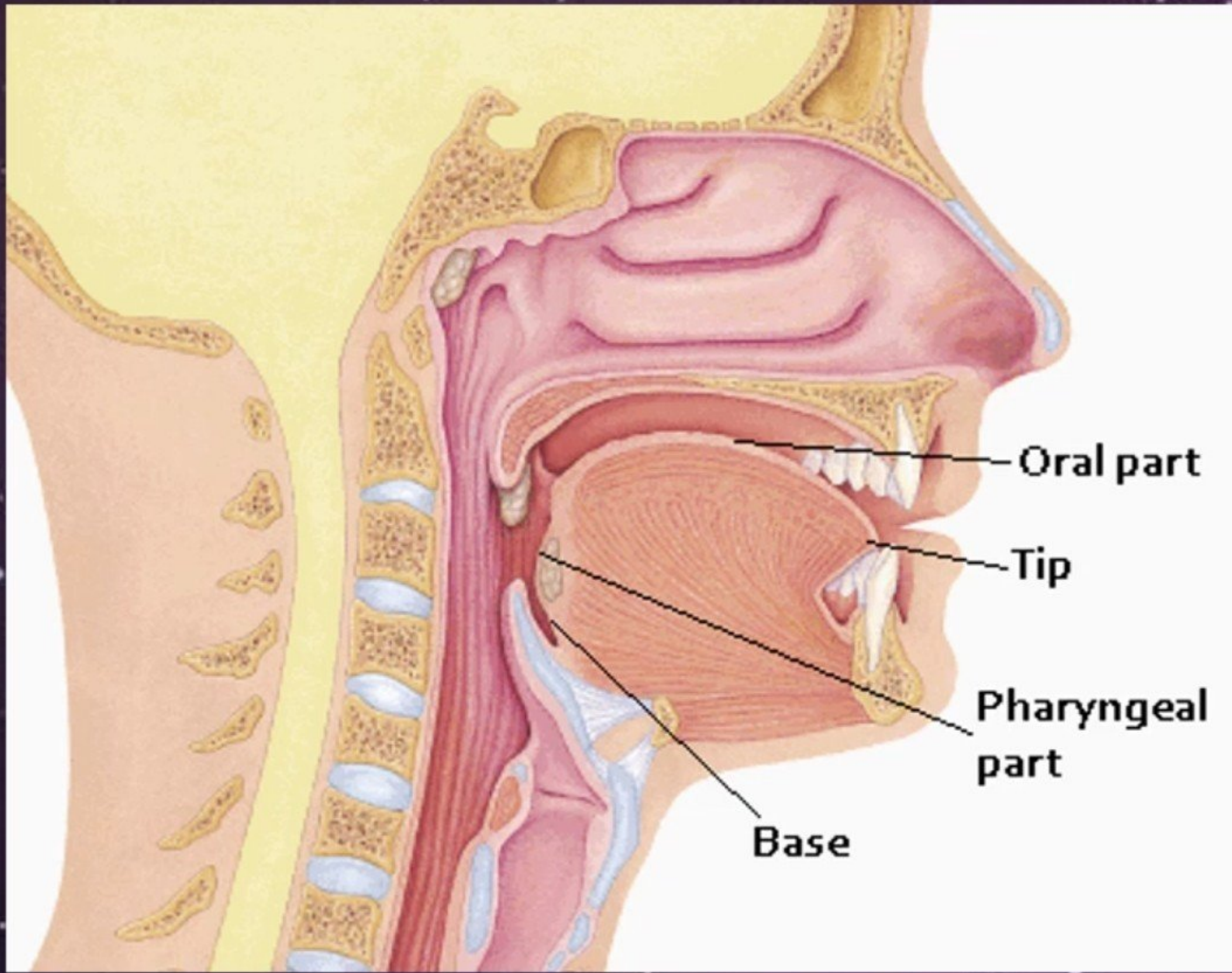


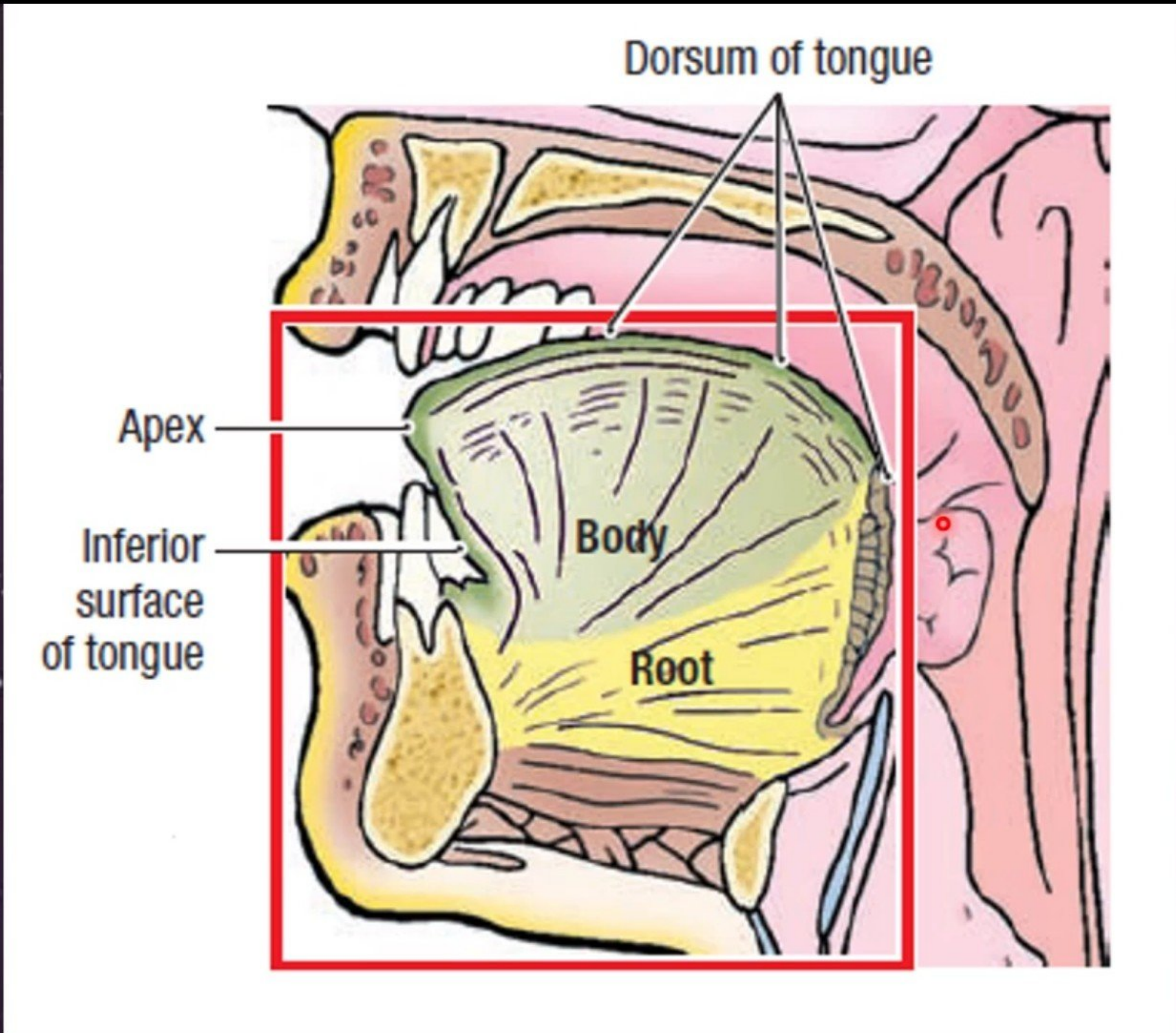

Central incisors
Lateral incisors
Canines
1st premolars
2nd premolars
1st molars
2nd molars
3rd molars
Incisive fossa
Palatine process of maxilla
Horizontal plate of palatine bone
Greater and lesser palatine foramina
Upper permanent teeth
Lower permanent teeth


Crown
Neck
Root
Enamel
Dentine and dentinal tubules
Interglobular spaces
Odontoblast layer
Interproximal spaces
Dental pulp containing vessels and nerves
Gingival (gum) epithelium (stratified)
Gingival groove
Lamina propria of gingiva (gum) (mandibular or maxillary periosteum)
Periodontium (alveolar periosteum)
Papilla
Cement
Root (central) canals containing vessels and nerves
Bone
Apical foramina

Tongue

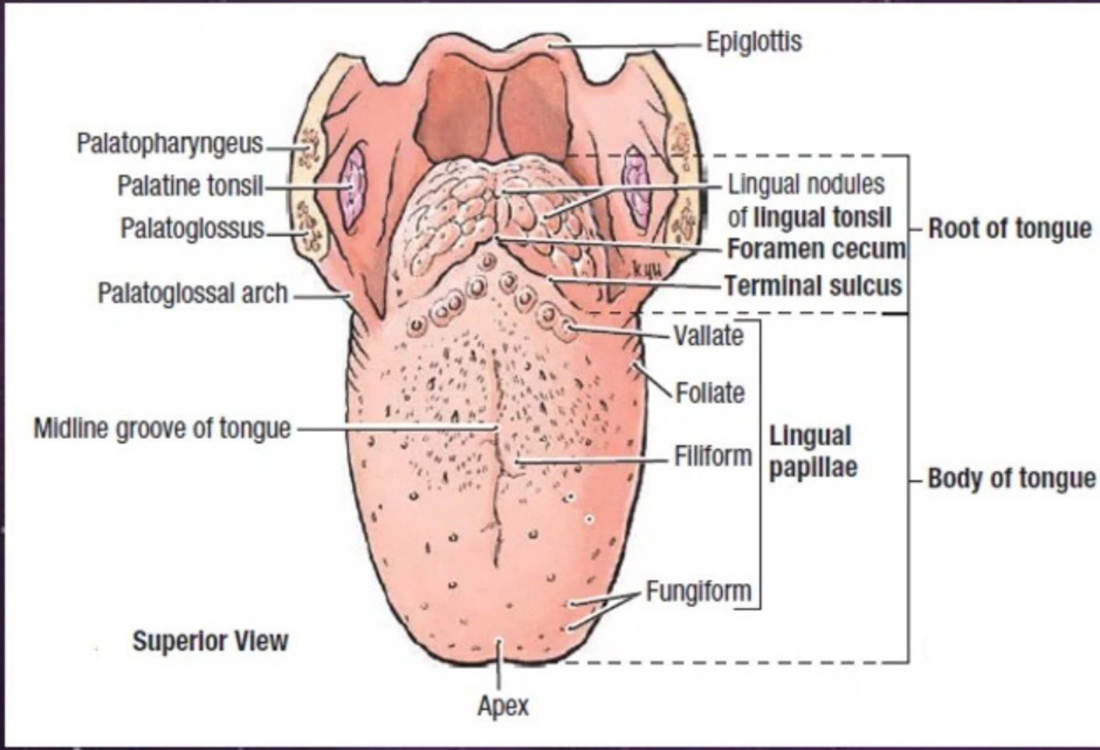
- Mass of striated muscles covered with the mucous membrane
- Divided into right and left halves by a median septum
- Three parts:
  - Oral (anterior ⅔)
  - Pharyngeal (posterior ⅓)
  - Root (base)
- Two surfaces:
  - Dorsal
  - Ventral




Dorsum of tongue
Apex
Inferior
surface
of tongue
Body
Root

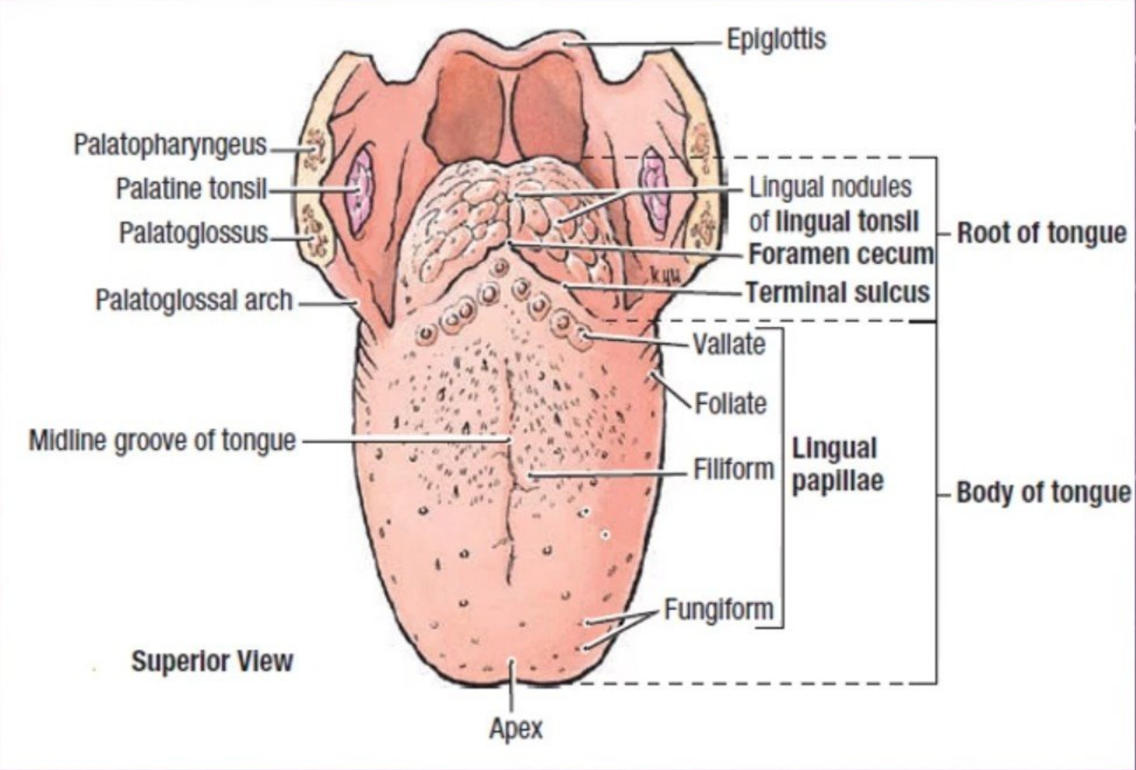
Dorsal Surface

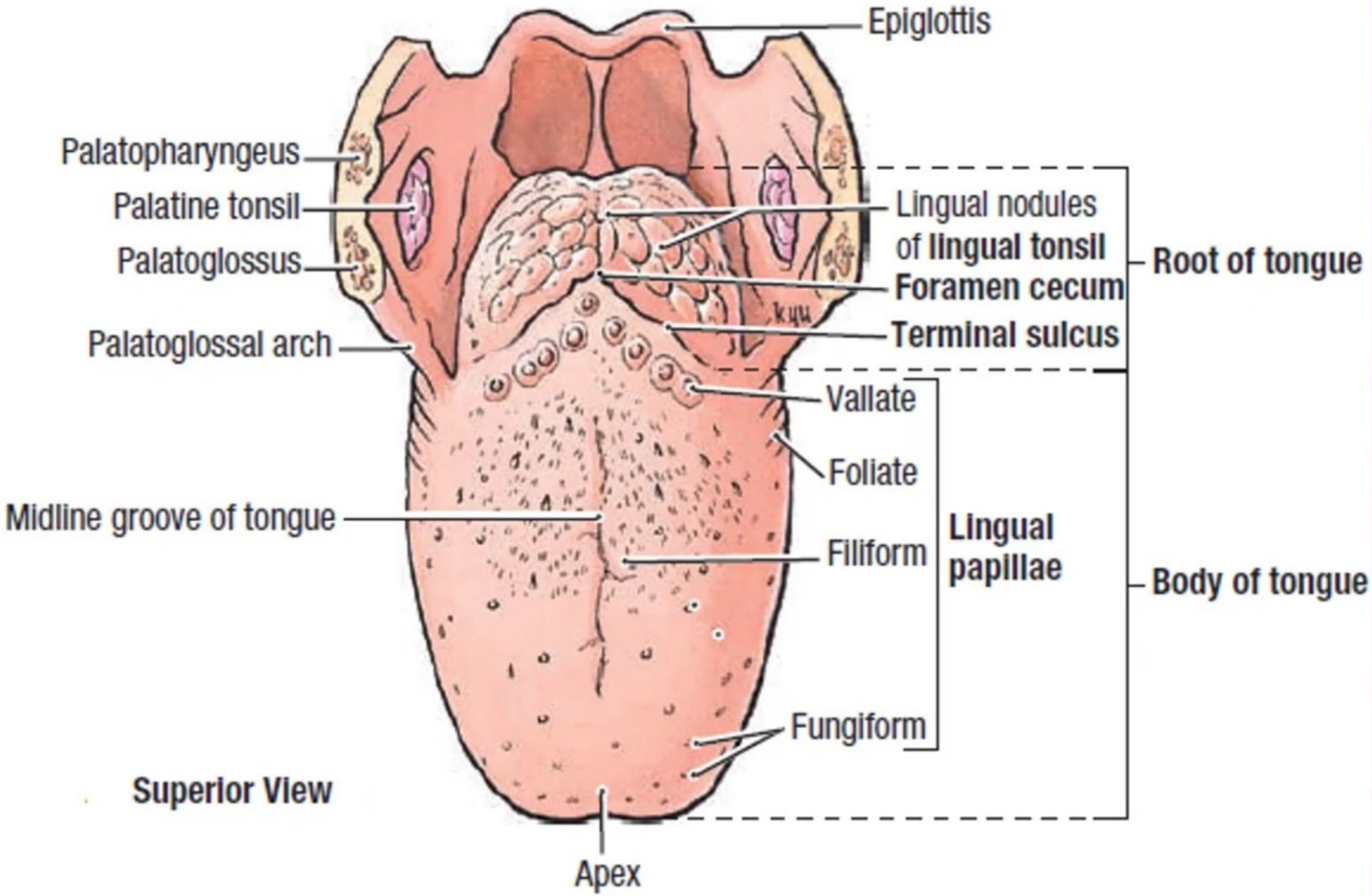
- Divided into anterior two third and posterior one third by a V-shaped sulcus terminalis.
- The apex of the sulcus faces backward and is marked by a pit called the foramen cecum
- Foramen cecum, an embryological remnant, marks the site of the upper end of the thyroglossal duct.

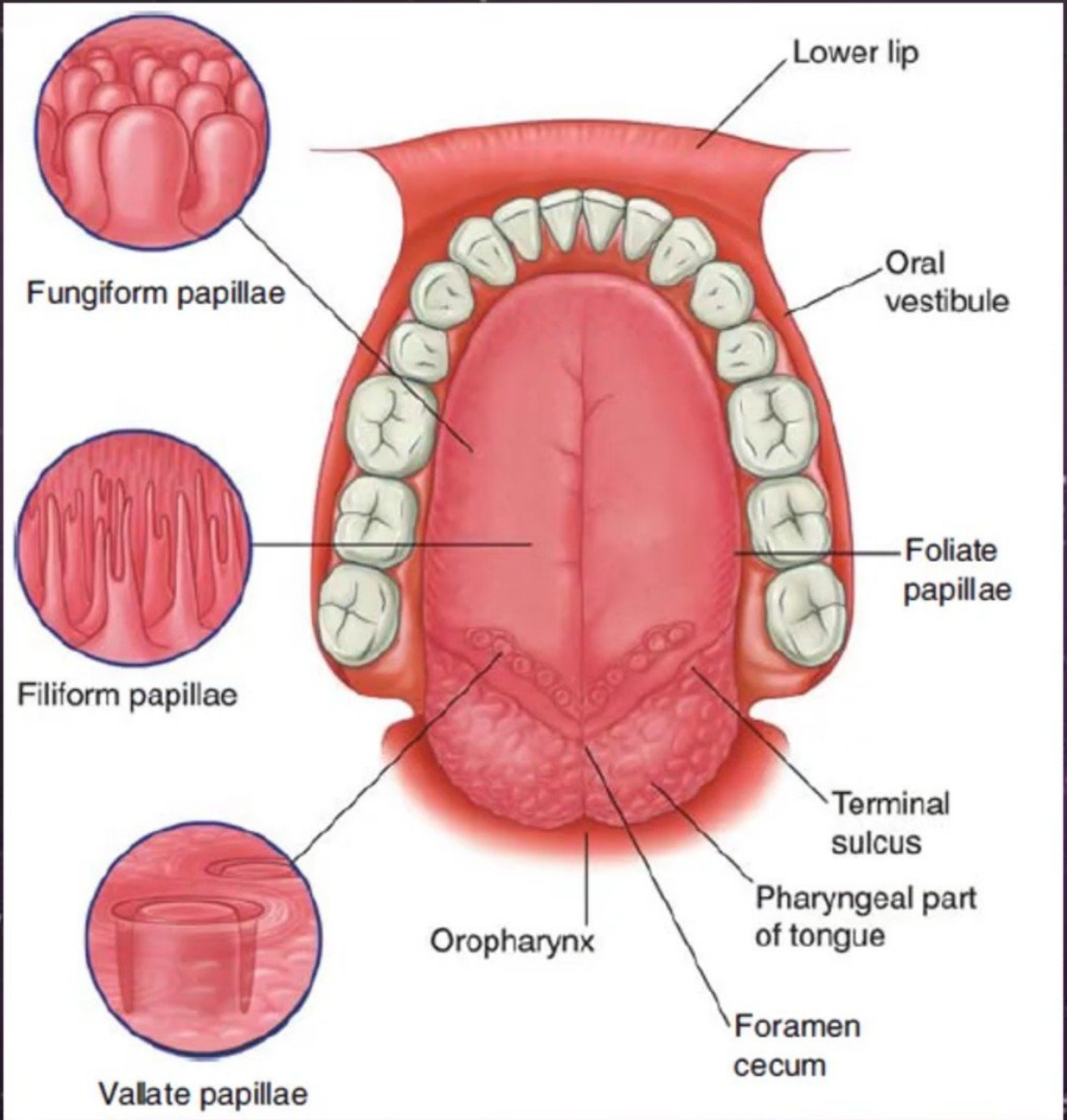


Dorsal Surface

- **Anterior two third**: mucosa is rough, shows three types of papillae:
  - Filliform
  - Fungiform
  - Vallate
- **Posterior one third**: No papillae but shows nodular surface because of underlying lymphatic nodules, the lingual tonsils.

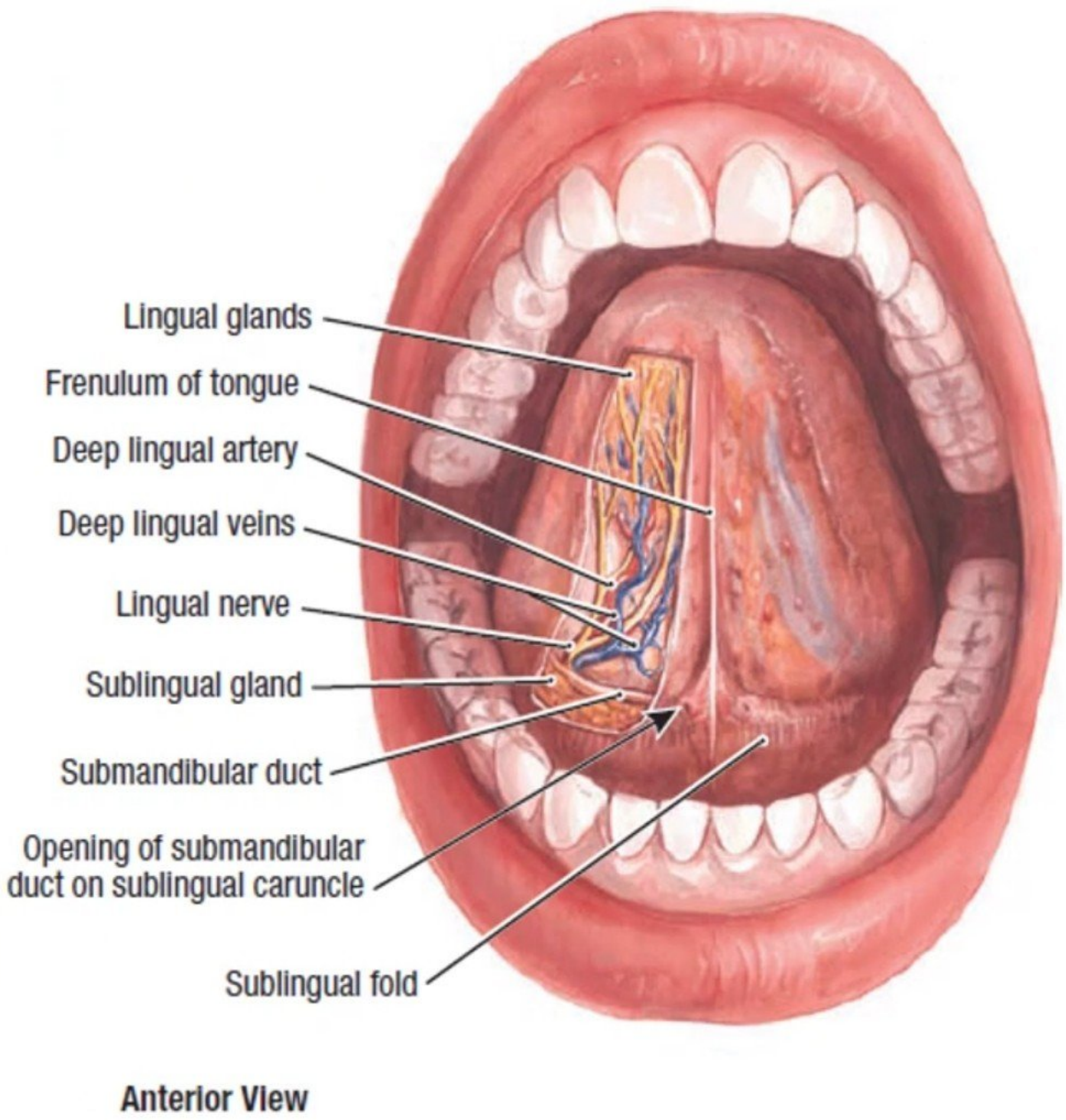



Epiglottis
Palatopharyngeus
Palatine tonsil
Palatoglossus
Palatoglossal arch
Lingual nodules of lingual tonsil
Foramen cecum
Terminal sulcus
Root of tongue
Vallate
Foliate
Filiform
Fungiform
Lingual papillae
Body of tongue
Midline groove of tongue
Superior View
Apex


Lower lip
Fungiform papillae
Oral vestibule
Filiform papillae
Foliate papillae
Terminal sulcus
Pharyngeal part of tongue
Oropharynx
Foramen cecum
Vallate papillae

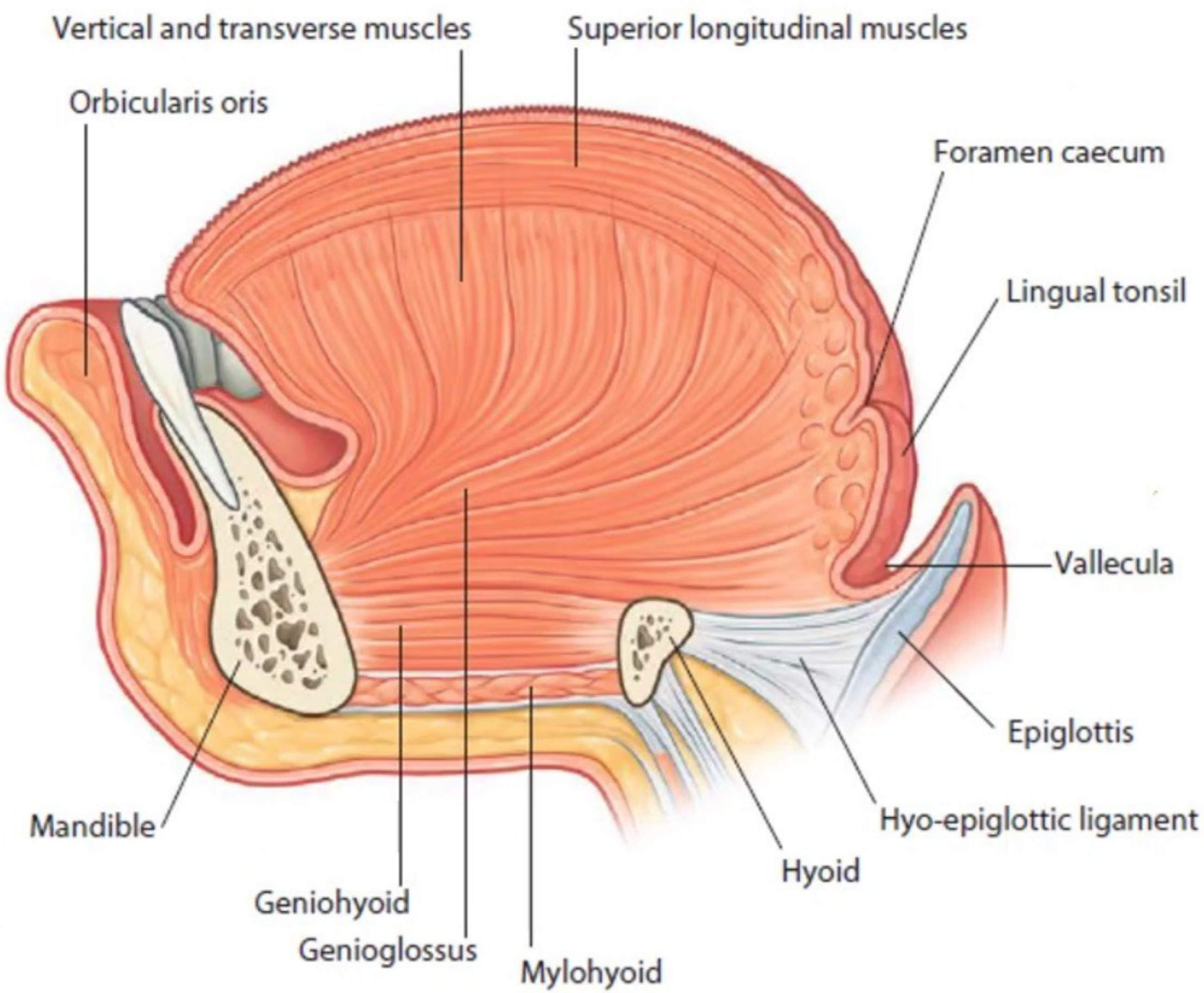
Ventral Surface

- Smooth (no papillae)
- In the midline anteriorly, a mucosal fold, frenulum connects the tongue with the floor of the mouth
- Lateral to frenulum, deep lingual vein can be seen through the mucosa
- Lateral to lingual vein, a fold of mucosa forms the plica fimbriata



Muscles

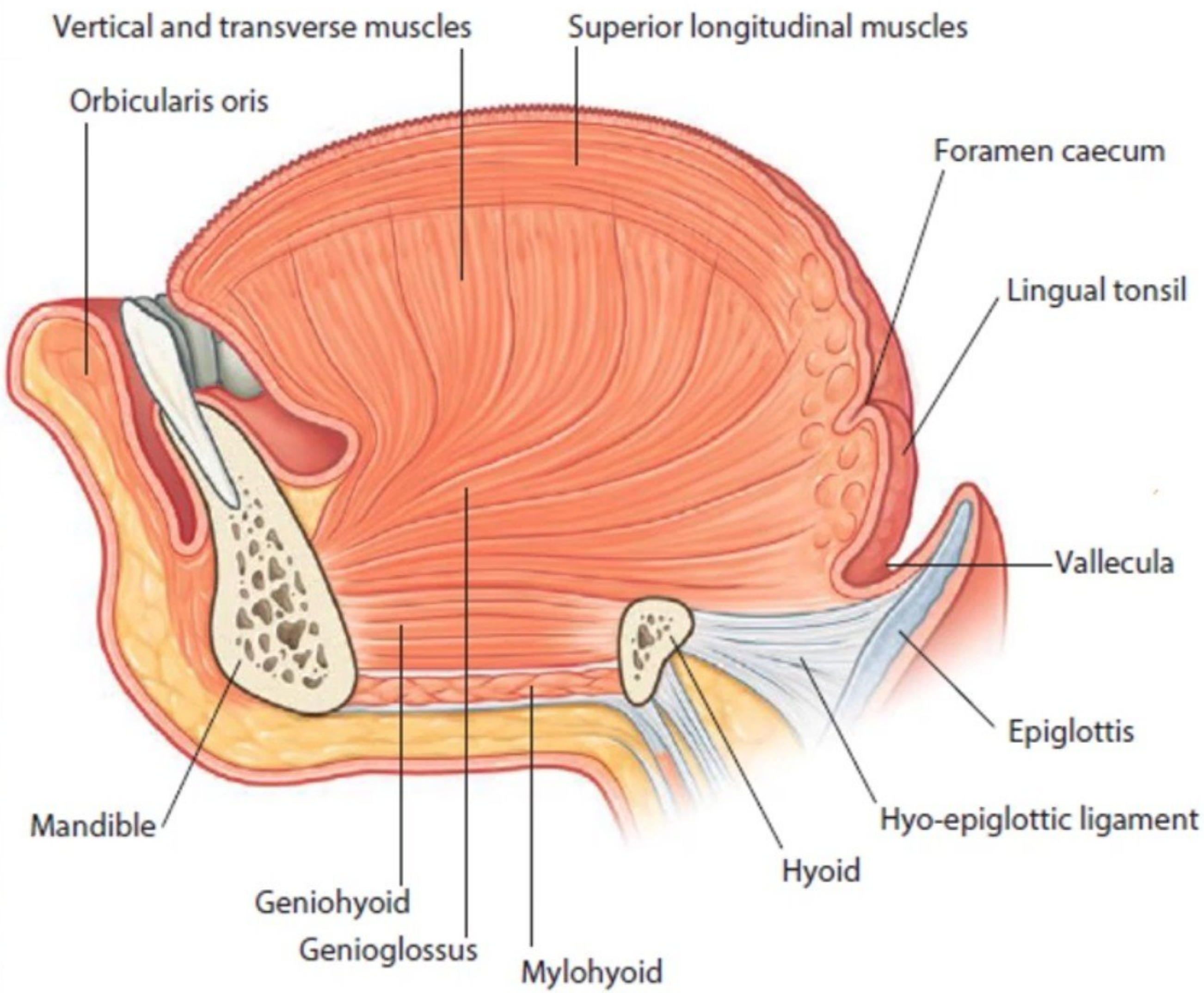
- The tongue is composed of two types of muscles:
  - Intrinsic
  - Extrinsic



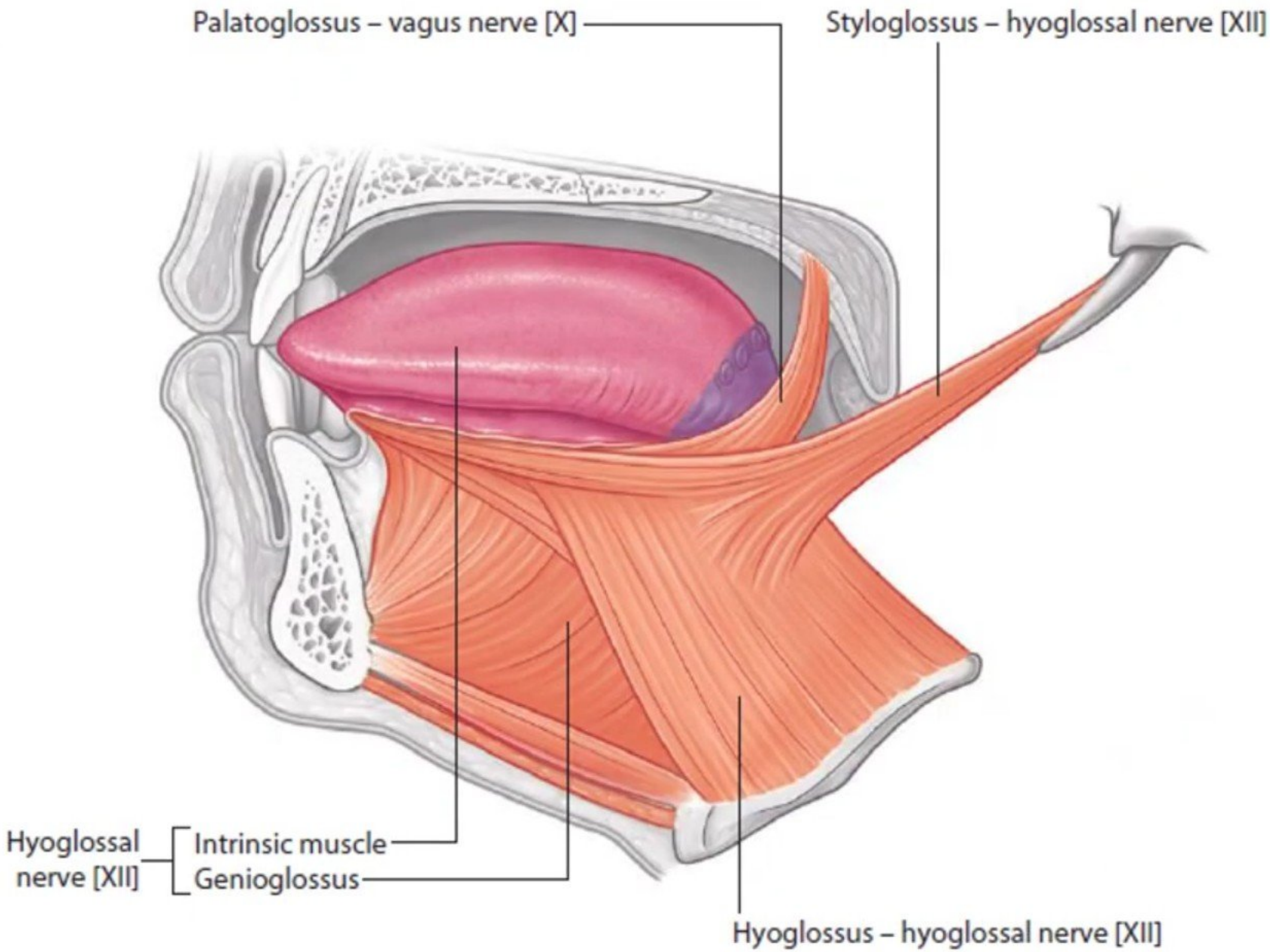
Paramedian sagittal section through tongue

Intrinsic Muscles

- Confined to tongue
- No bony attachment
- Consist of:
  - Longitudinal fibers
  - Transverse fibers
  - Vertical fibers
- Function: Alter the shape of the tongue.



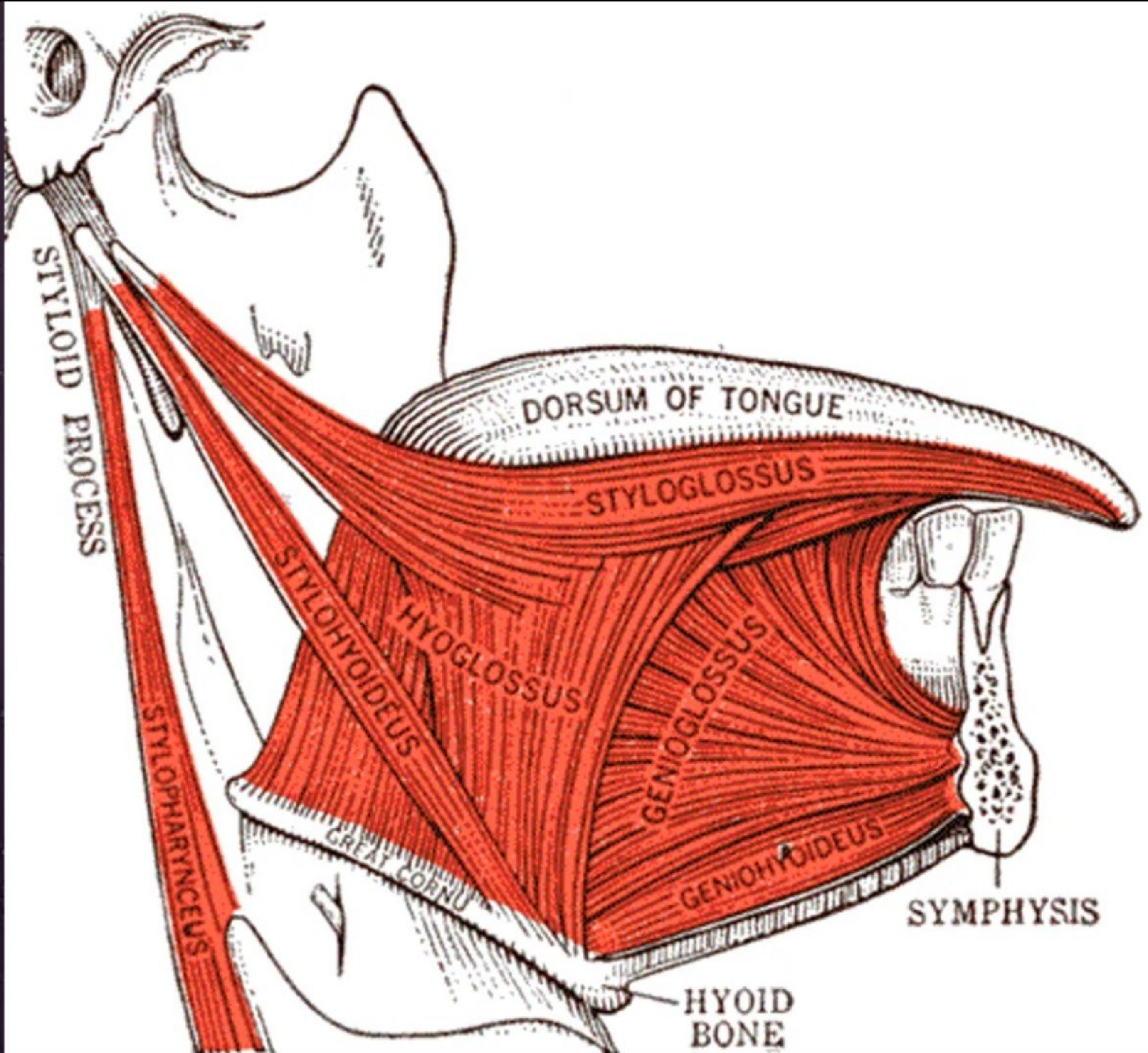
Paramedian sagittal section through tongue



Innervation of the tongue

Extrinsic Muscles

- Connect the tongue to the surrounding structures: the soft palate and the bones (mandible, hyoid bone, styloid process)
- Include:
  - Palatoglossus
  - Genioglossus
  - Hyoglossus
  - Styloglossus
- Function: Help in movements of the tongue

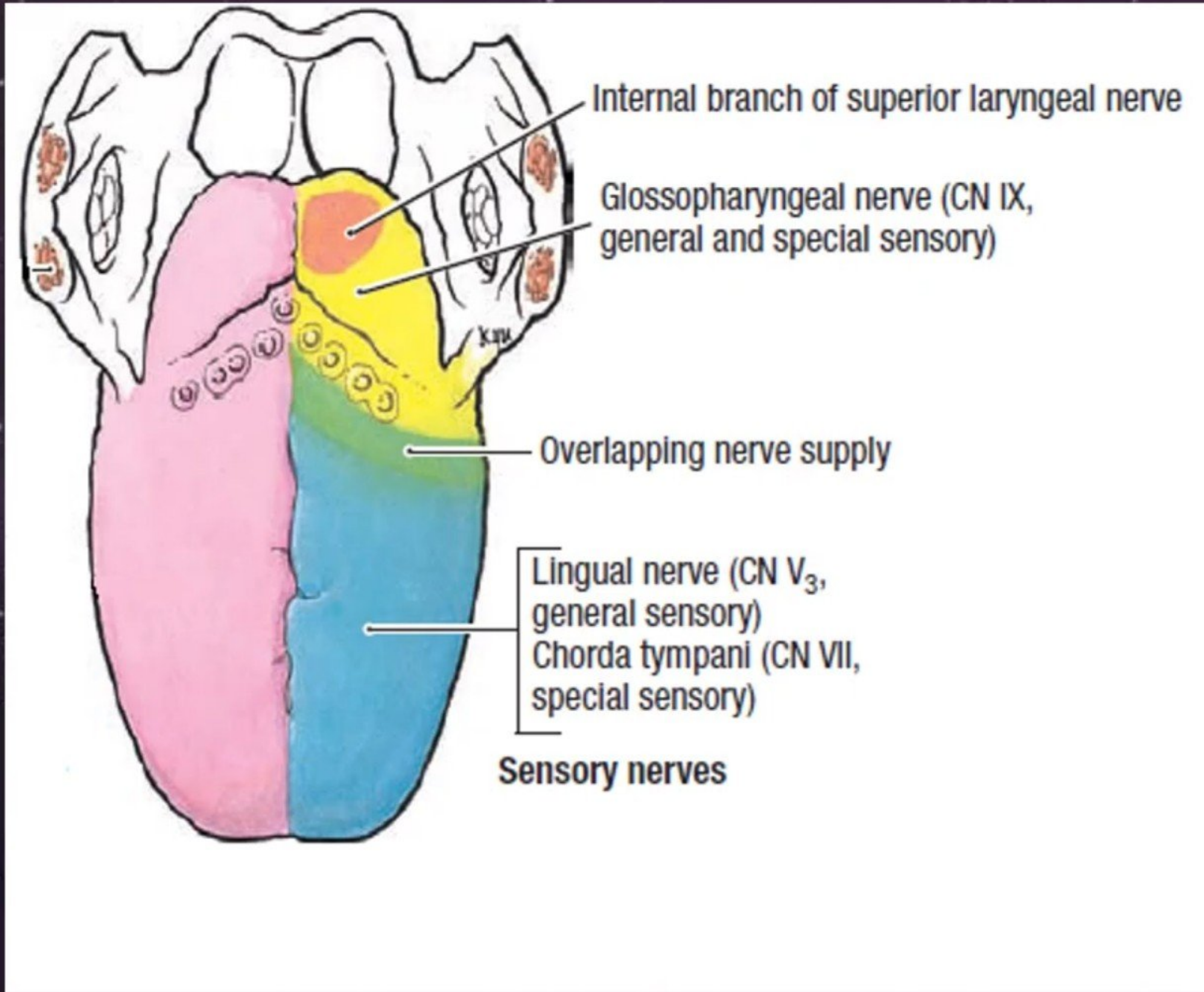


Movements

- **Protrusion:**
  - Genioglossus on both sides acting together
- **Retraction:**
  - Styloglossus and hyoglossus on both sides acting together
- **Depression:**
  - Hyoglossus and genioglossus on both sides acting together
- **Elevation:**
  - Styloglossus and palatoglossus on both sides acting together

Sensory Nerve Supply

- **Anterior ⅔:**
  - General sensations: Lingual nerve
  - Special sensations : chorda tympani
- **Posterior ⅓:**
  - General & special sensations: glossopharyngeal nerve
- **Base:**
  - General & special sensations: internal laryngeal nerve



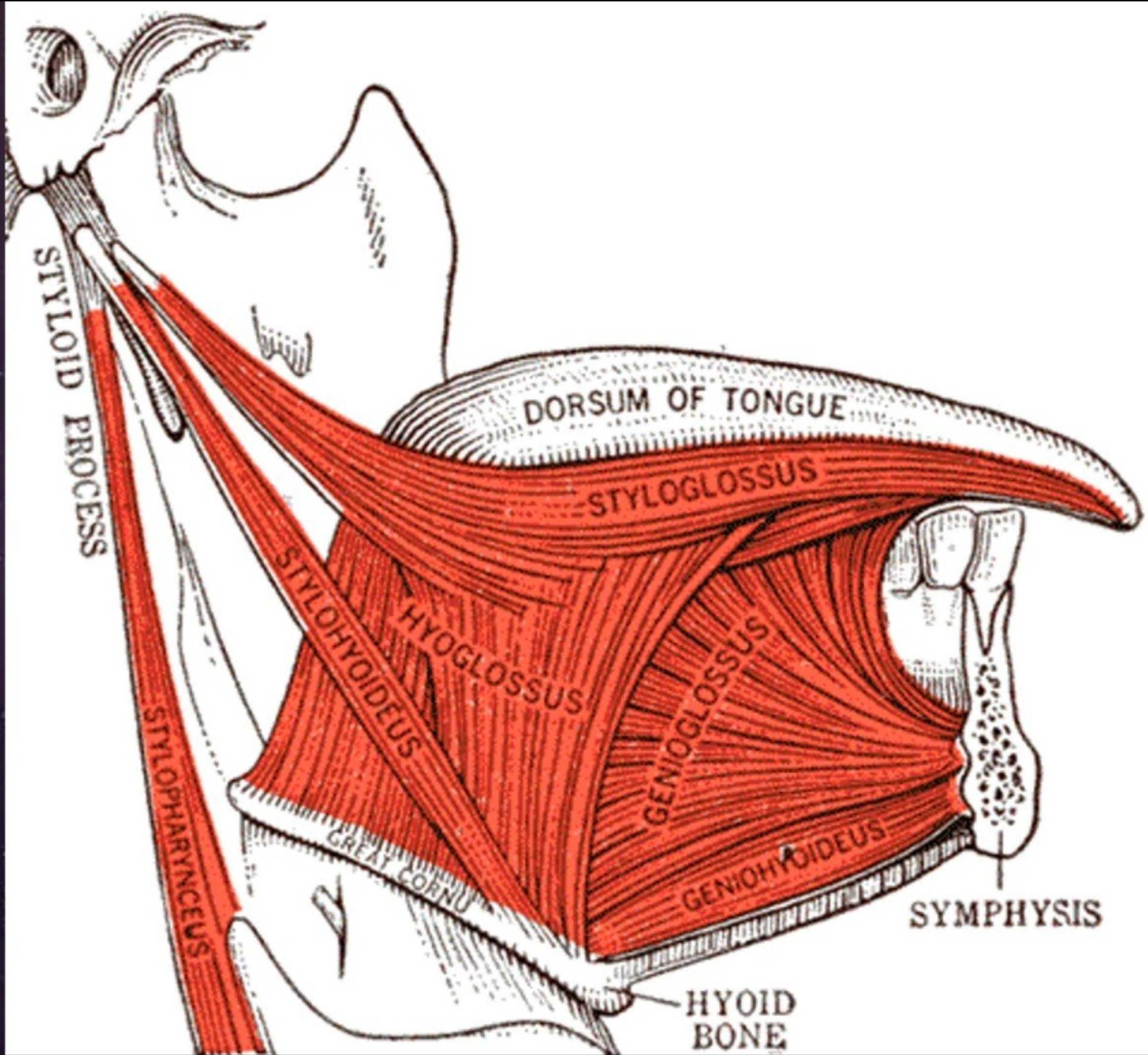
Sensory

Anterior two-thirds (oral)
- general sensation
  Mandibular nerve [V_3] via lingual nerve
- special sensation (taste)
  Facial nerve [VII] via chorda tympani

Posterior one-third (pharyngeal)
- general and special sensation (taste)
  Glossopharyngeal nerve [IX]

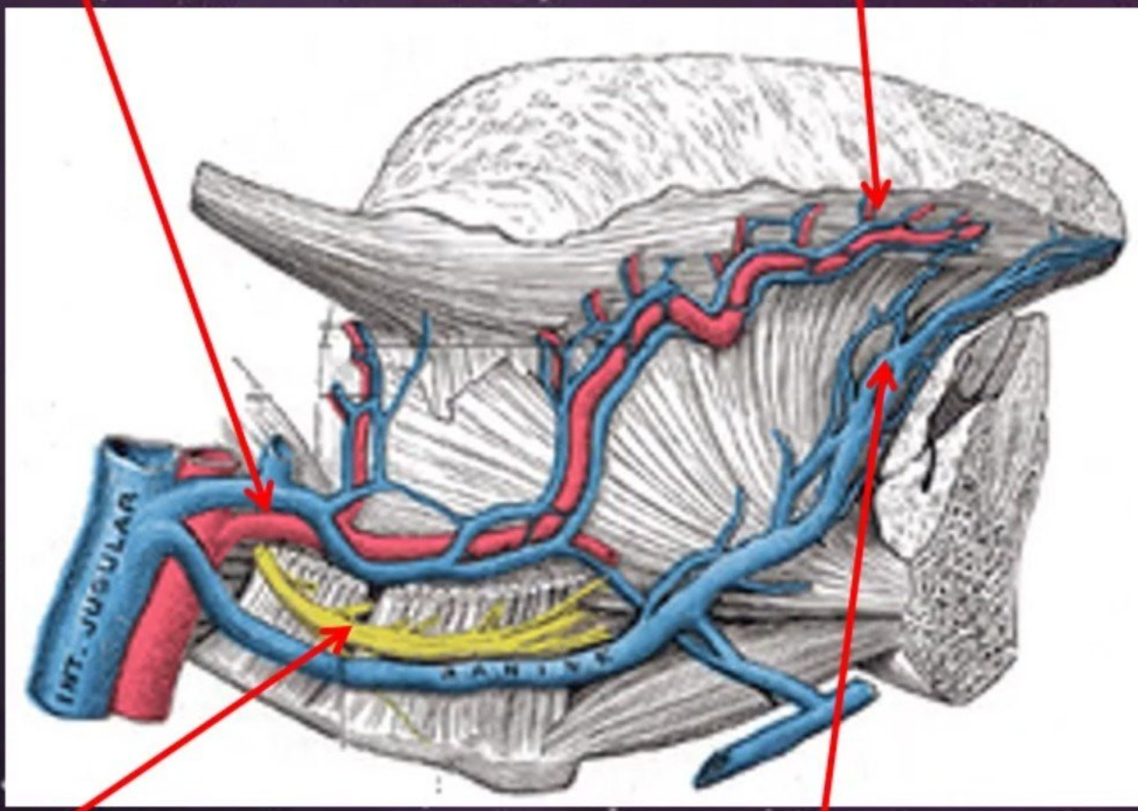
Motor Nerve Supply

- **Intrinsic muscles:**
  - Hypoglossal nerve.
- **Extrinsic muscles:**
  - All supplied by the hypoglossal nerve, except the palatoglossus.
- The palatoglossus supplied by the pharyngeal plexus



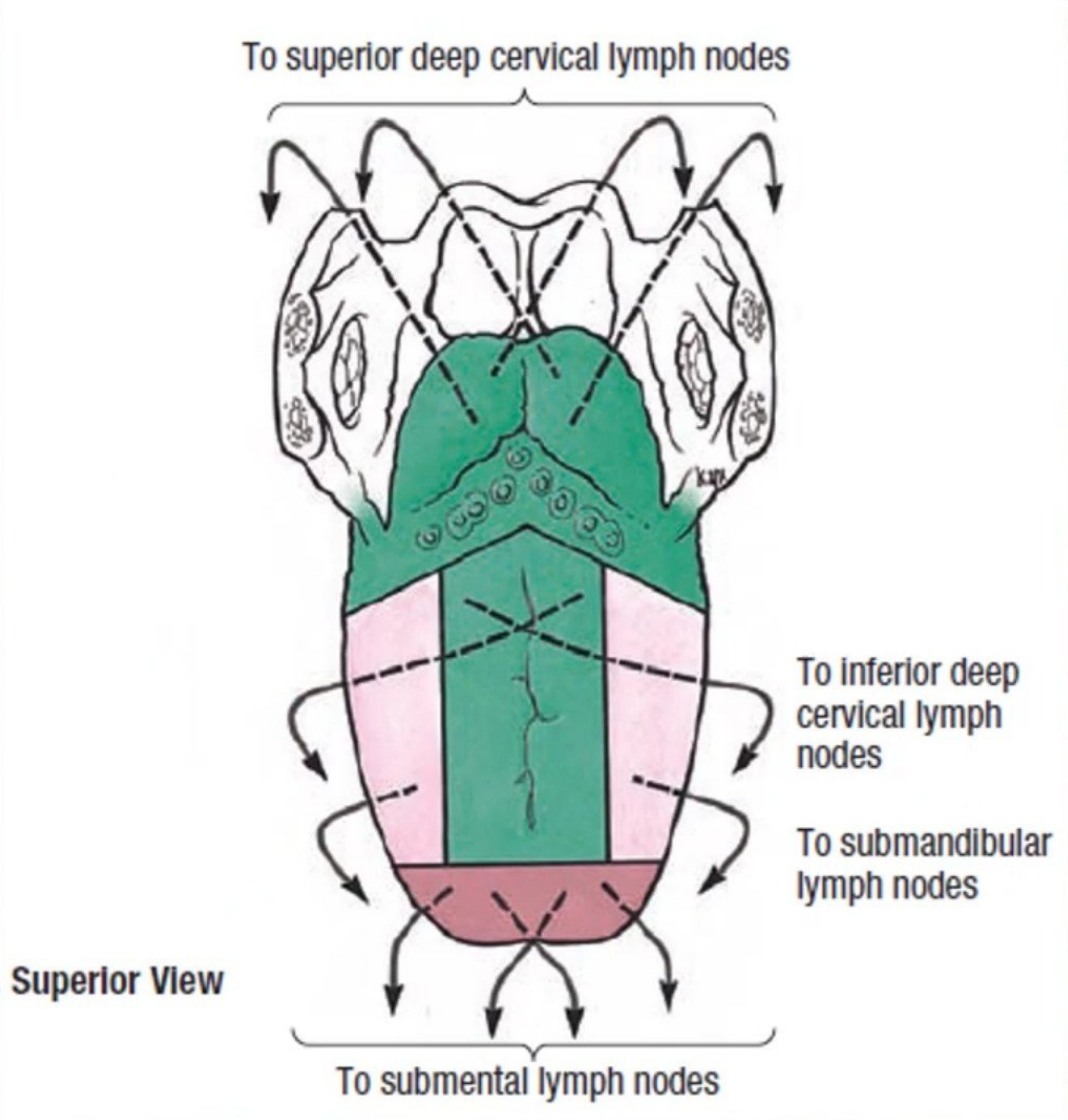
Blood Supply

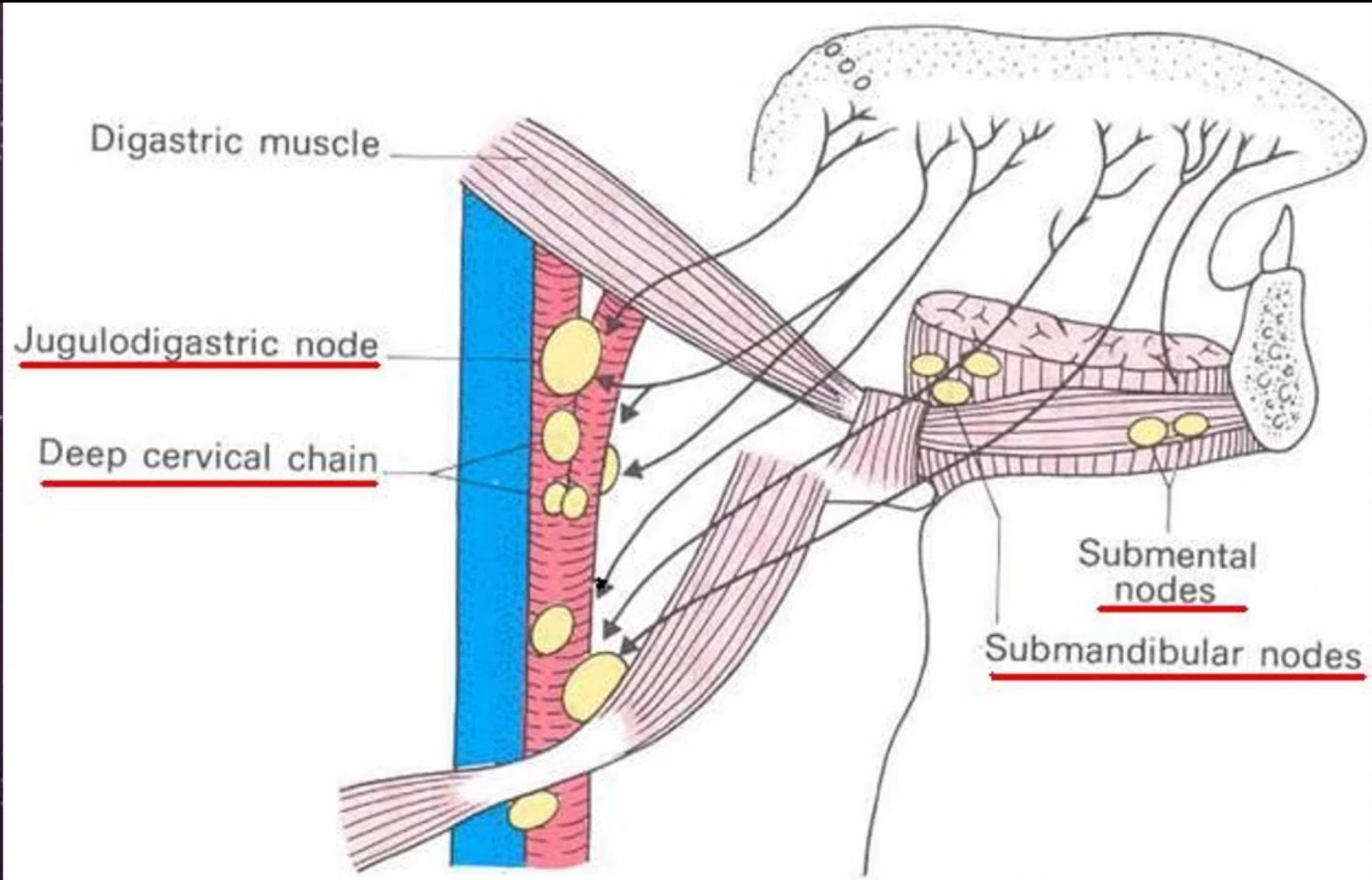
- Arteries:
  - Lingual artery
  - Tonsillar branch of facial artery
  - Ascending pharyngeal artery
- Veins:
  - Lingual vein, ultimately drains into the internal jugular vein



Lymphatic Drainage

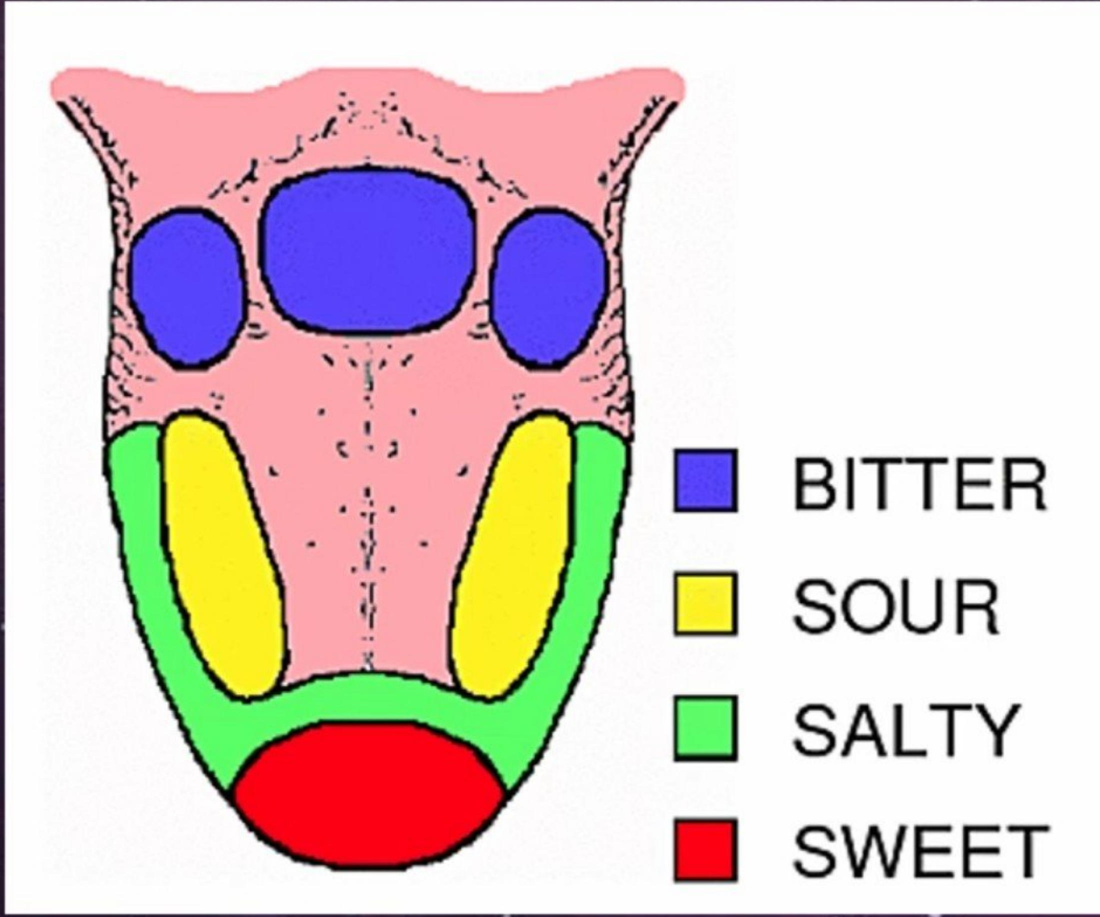
- **Tip:**
  - Submental nodes bilaterally & then deep cervical nodes.
- **Anterior two third:**
  - Submandibular unilaterally & then deep cervical nodes.
- **Posterior third:**
  - Deep cervical nodes (jugulodigastric mainly).




Digastric muscle
Jugulodigastric node
Deep cervical chain
Submental nodes
Submandibular nodes

Functions

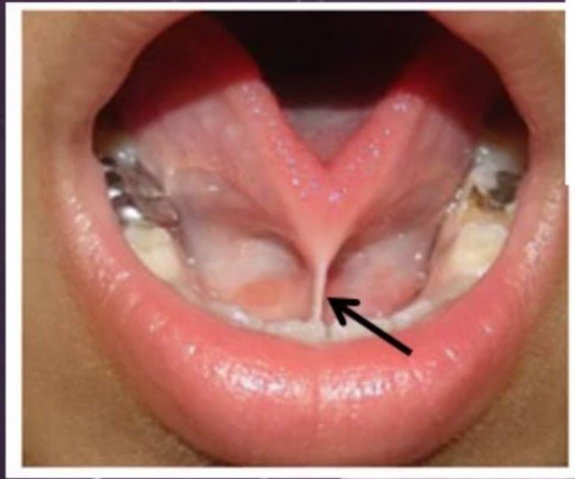
- The tonge is the most important articulator for **speech production**. During speech, the tongue can make amazing range of movements
- The primary function of the tongue is to provide a mechanism for **taste.** Taste buds are located on different areas of the tongue, but are generally found around the edges. They are sensitive to four main tastes: Bitter, Sour, Salty & Sweet.



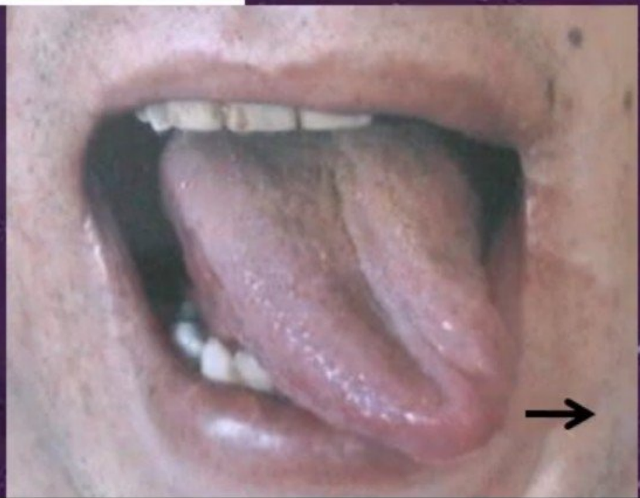
- The tongue is needed for sucking, chewing, swallowing, eating, drinking, kissing, sweeping the mouth for food debris and other particles and for making funny faces (poking the tongue out, waggling it)
- Trumpeters and horn & flute players have very well developed tongue muscles, and are able to perform rapid, controlled movements or articulations

Clinical Notes

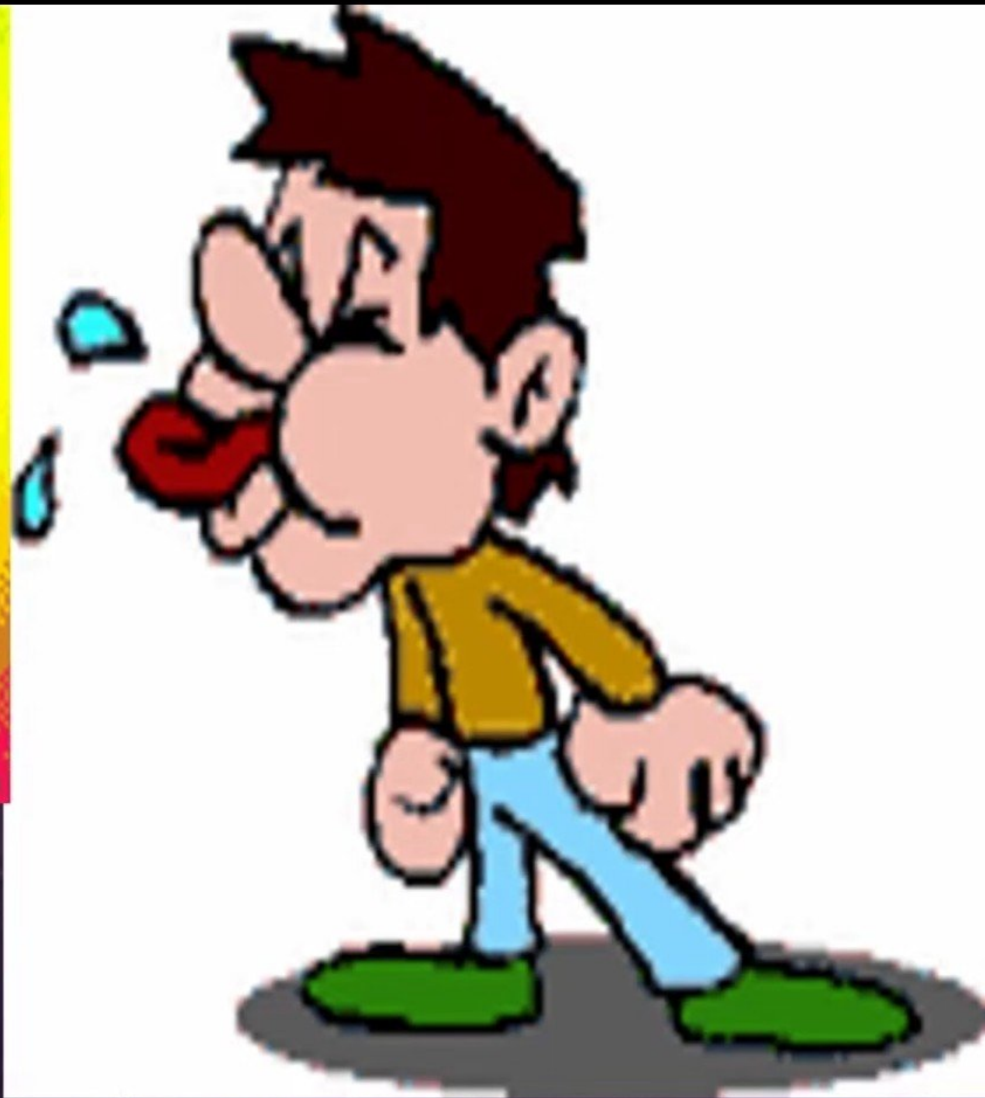
- Lacerations of the tongue
- Tongue-Tie (ankyloglossia) (due to large frenulum)



- Lesion of the hypoglossal nerve
  - The protruded tongue deviates toward the side of the lesion
  - Tongue is atrophied & wrinkled.



Anger is a condition in which the tongue works faster than the mind.



‘If there is goodness in your heart, it will come to your tongue’.

Will be continue …